Computer Vision —— Object Recognition

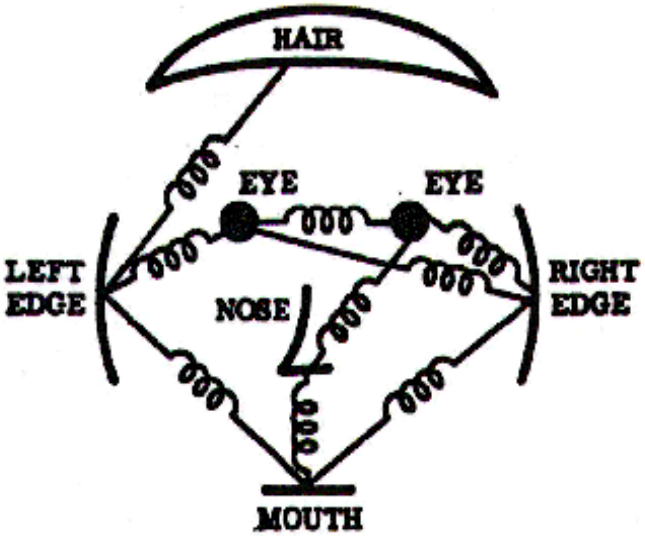
2026/10/13

桑 农　王岳环　高常鑫

Slides from CV course with R. Fergus, and S. Savarese, S. Lazebnik, D. Lowe, P. Viola

Agenda

- Introduction
- Objects & Faces recognition
- Bag-of-words models
- Part-based models



Part-based models

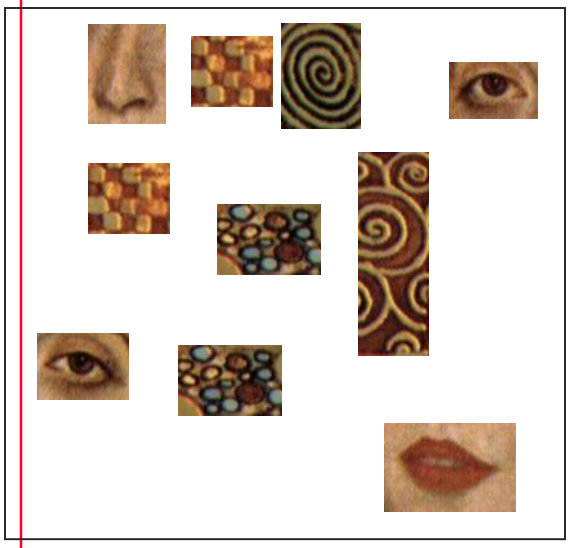
Overview

- Representation
  - Location
  - Appearance
  - Generative interpretation
- Learning
- Distance transforms
- Other approaches using parts
- Felzenszwalb, McAllester, Ramanan CVPR2008 / Felzenszwalb, Girshick, McAllester, Ramanan TPAMI2010

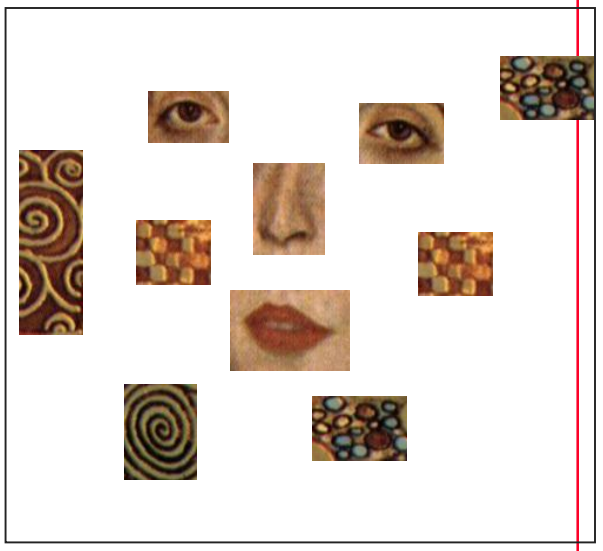
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Problem with bag-of-words







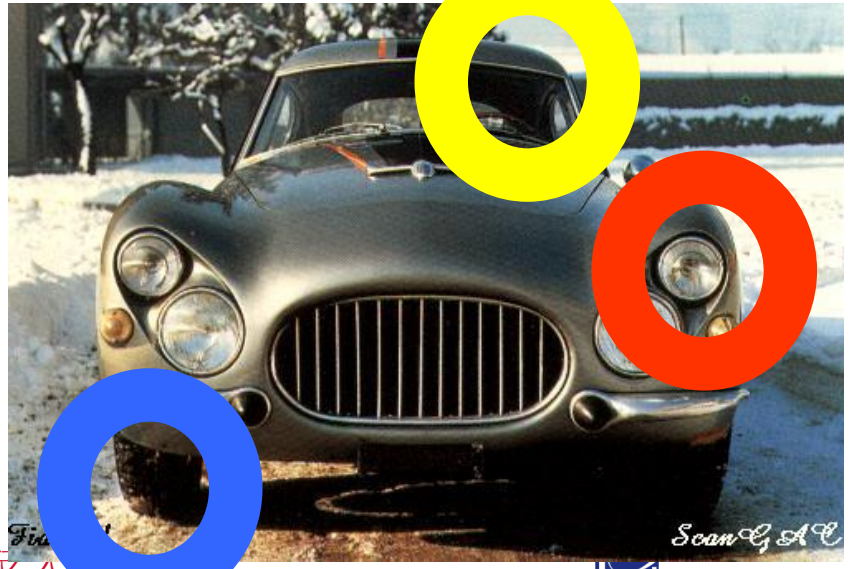
- All have equal probability for bag-of-words methods
- Location information is important

Model: Parts and Structure









Representation

- Object as set of parts
  - Generative representation
- Model:
  - Relative locations between parts
  - Appearance of part
- Issues:
  - How to model location
  - How to represent appearance
  - Sparse or dense (pixels or regions)
  - How to handle occlusion/clutter

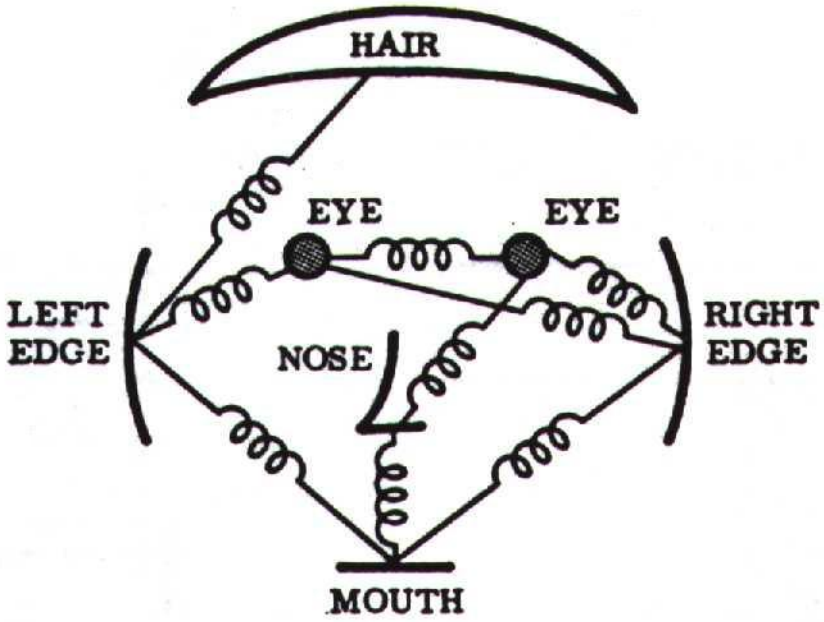
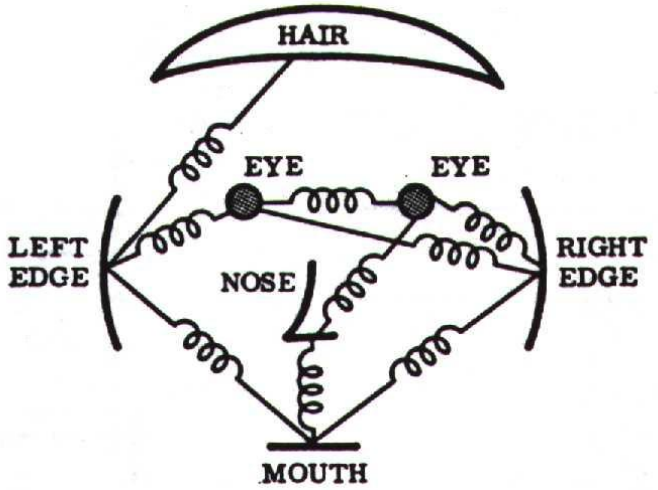


Figure from [Fischler & Elschlager 73]

History of Parts and Structure approaches

- Fischler & Elschlager 1973

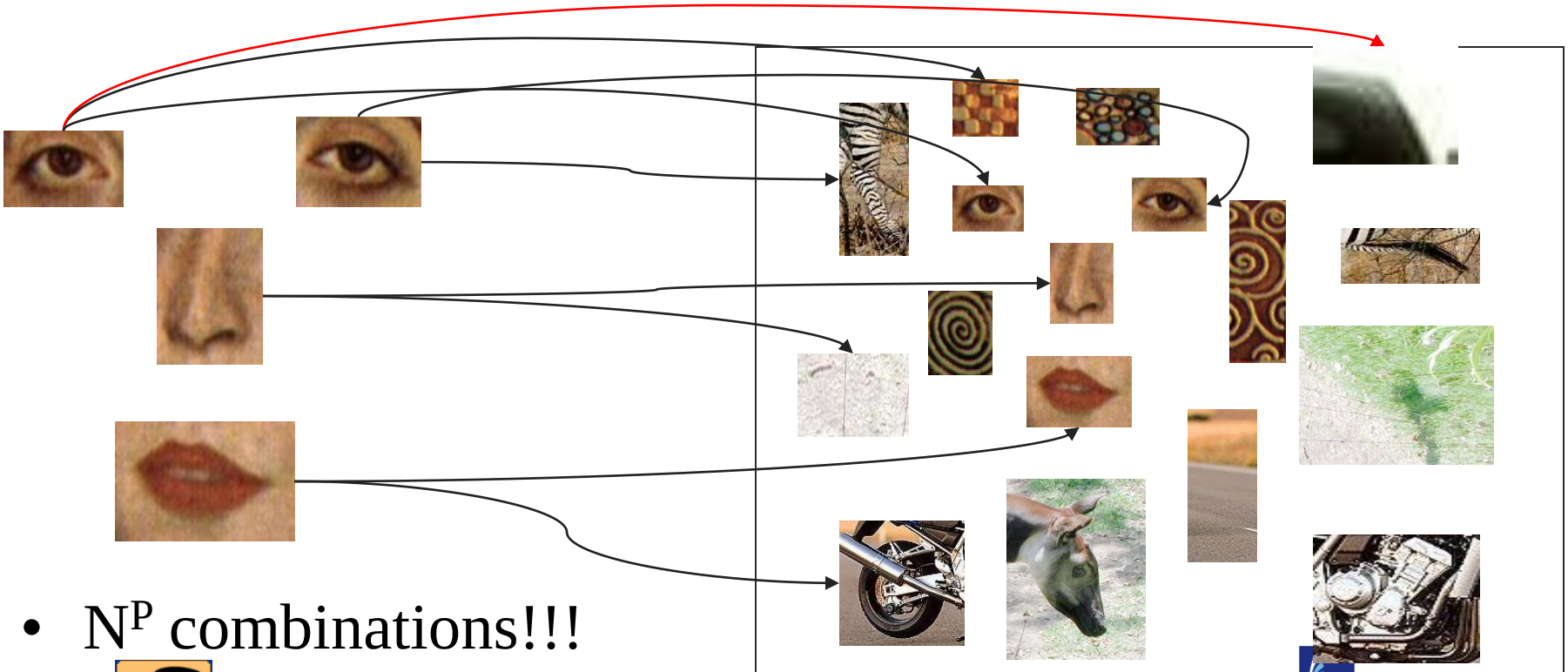


- Yuille ‘91
- Brunelli & Poggio ‘93
- Lades, v.d. Malsburg et al. ‘93
- Cootes, Lanitis, Taylor et al. ‘95
- Amit & Geman ‘95, ‘99
- Perona et al. ‘95, ‘96, ’98, ’00, ’03, ‘04, ‘05
- Felzenszwalb & Huttenlocher ’00, ’04
- Crandall & Huttenlocher ’05, ’06
- Leibe & Schiele ’03, ’04

- Many papers since 2000

The correspondence problem

- Model with P parts
- Image with N possible locations for each part



- N^P combinations!!!

Sparse representation

+ Computationally tractable (10^5 pixels → 10^1 -- 10^2 parts)
+ Generative representation of class
+ Avoid modeling global variability
+ Success in specific object recognition



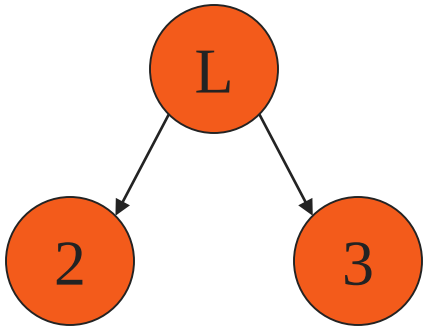


- Throw away most image information
- Parts need to be distinctive to separate from other classes

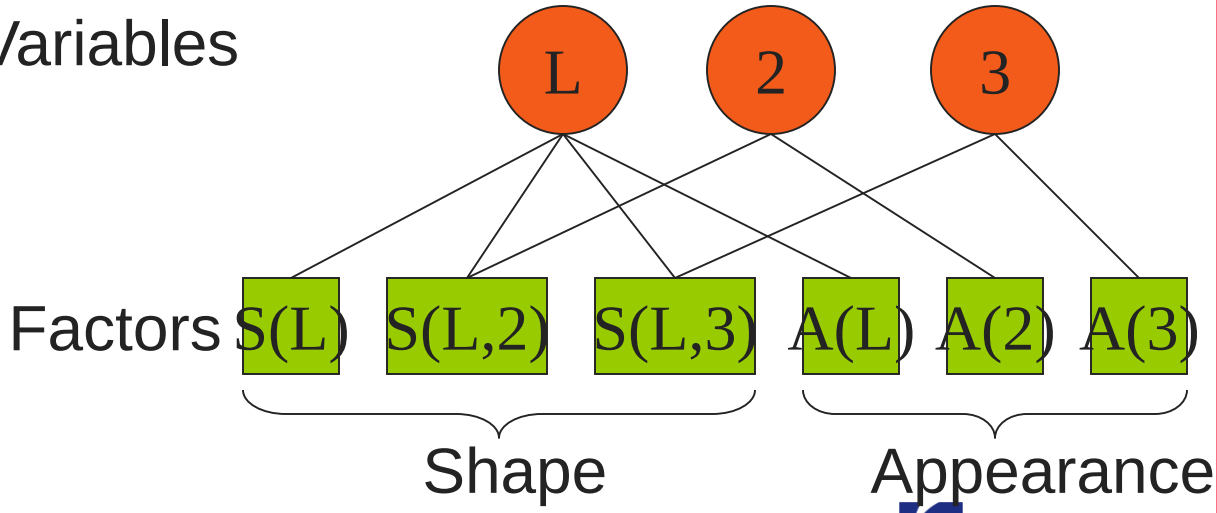
Connectivity of parts

- Complexity is given by size of **maximal clique** in graph
- Consider a 3 part model
  - Each part has set of N possible locations in image
  - Location of parts 2 & 3 is independent, given location of L
  - Each part has an appearance term, independent between parts.

Shape Model



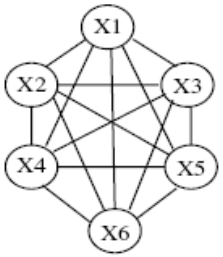
Factor graph



Different connectivity structures

Fergus et al. ’03
Fei-Fei et al. ‘03

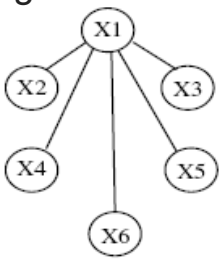
$O(N^6)$



a) Constellation [13]

Crandall et al. ‘05
Fergus et al. ’05

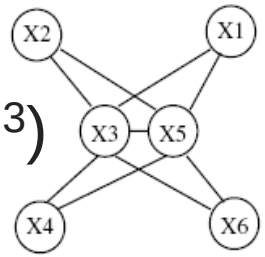
$O(N^2)$



b) Star shape [9, 14]

Crandall et al. ‘05

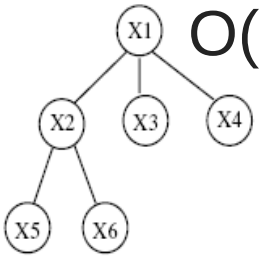
$O(N^3)$



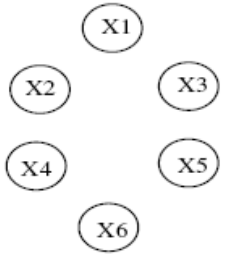
c) k-fan ($k = 2$) [9]

Felzenszwalb & Huttenlocher ‘00

$O(N^2)$

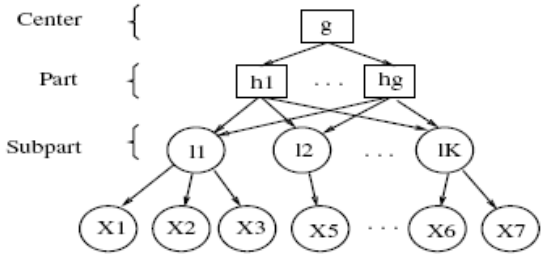


d) Tree [12]



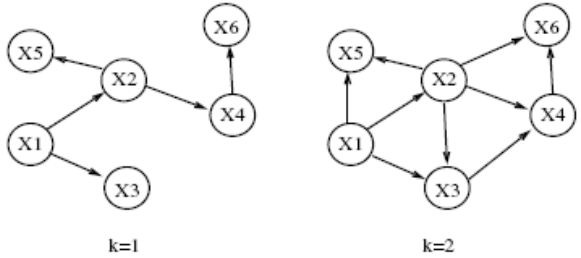
e) Bag of features [10, 21]

Csurka ’04
Vasconcelos ‘00



f) Hierarchy [4]

Bouchard & Triggs ‘05



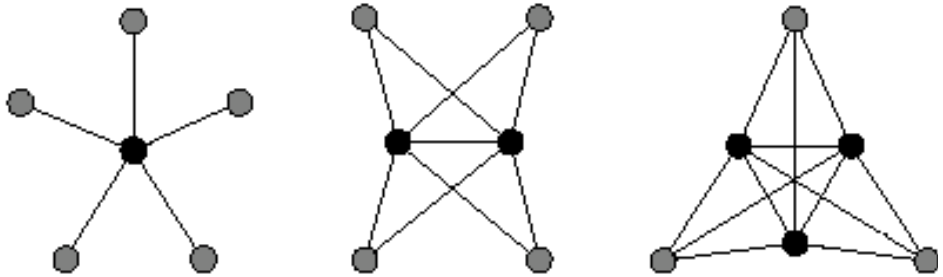
g) Sparse flexible model

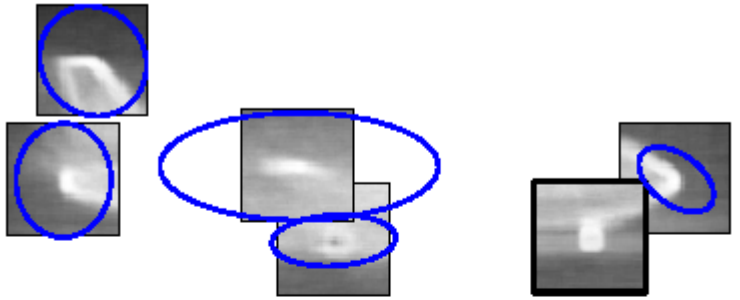
Carneiro & Lowe ‘06

from Sparse Flexible Models of Local Features
Gustavo Carneiro and David Lowe, ECCV 2006

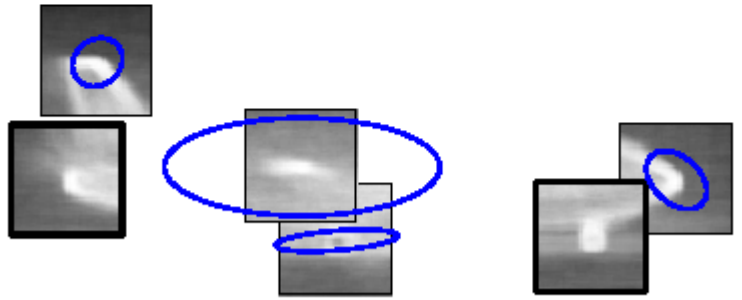
How much does shape help?

- Crandall, Felzenszwalb, Huttenlocher CVPR'05
- Shape variance increases with increasing model complexity
- Do get some benefit from shape

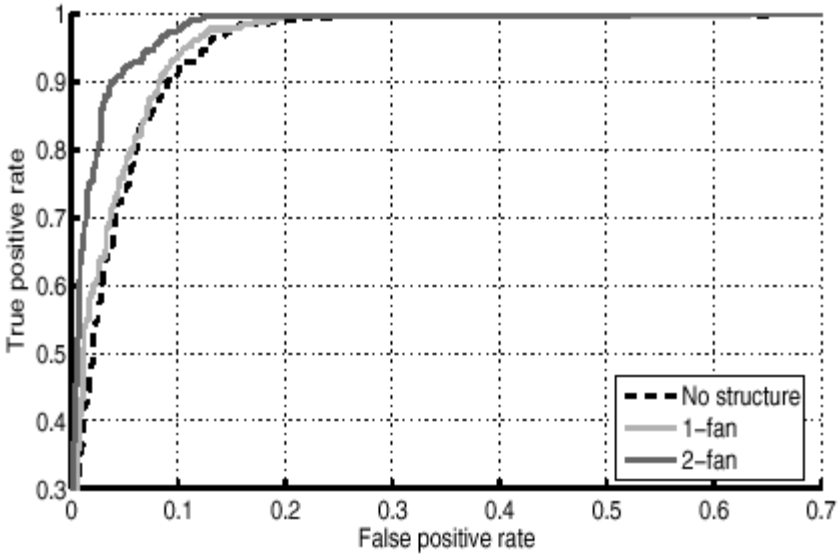




(a) Airplane, 1-fan

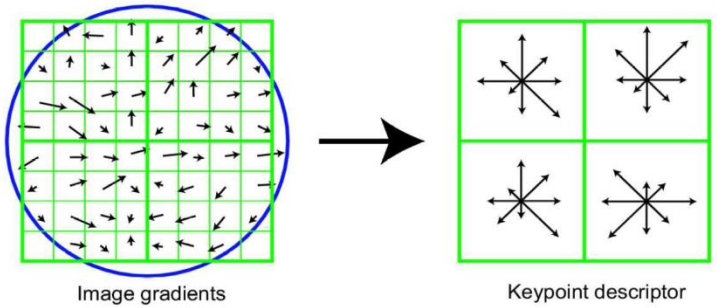


(b) Airplane, 2-fan

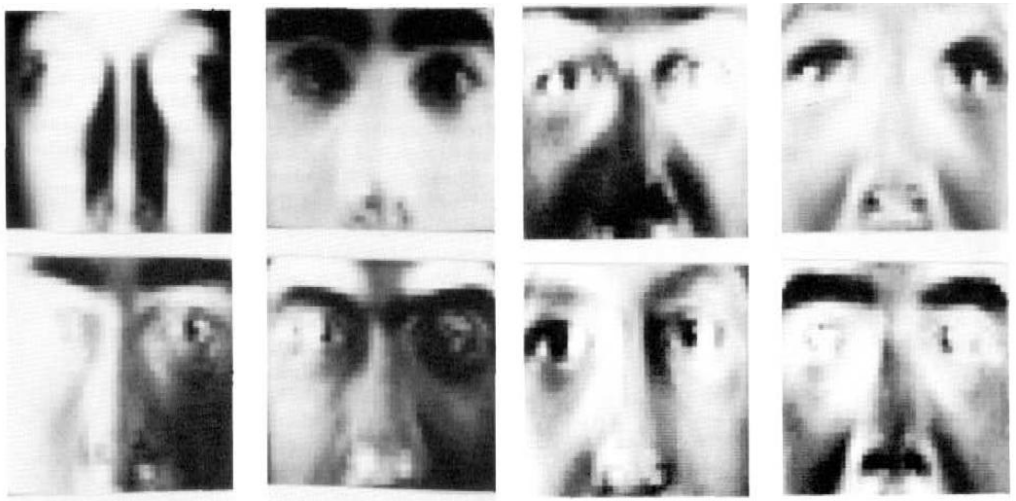


Appearance representation

- SIFT



- PCA



- Decision trees

[Lepetit and Fua CVPR 2005]

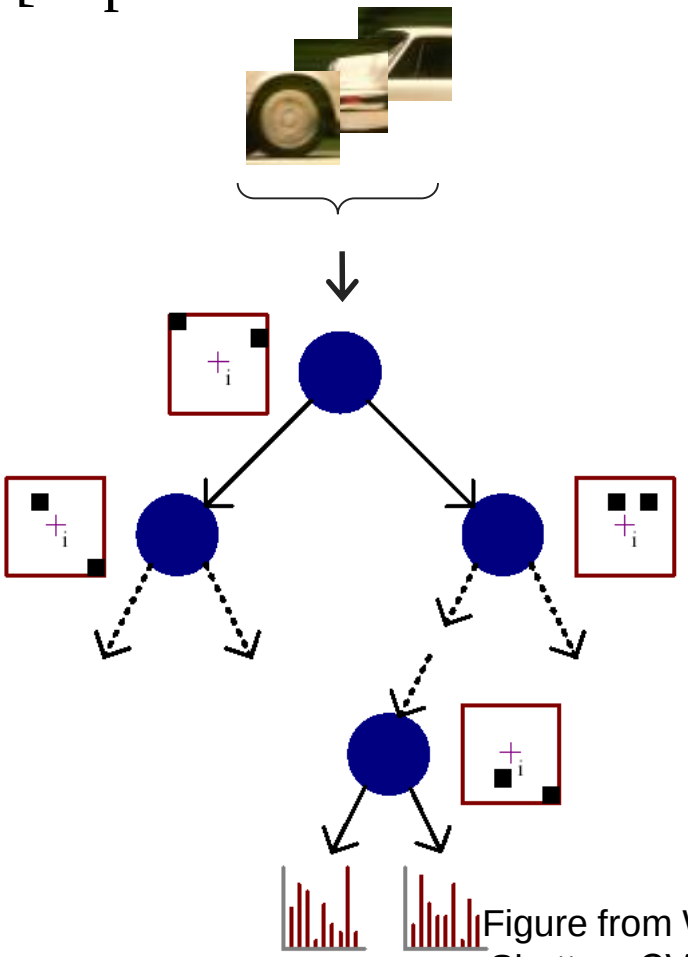


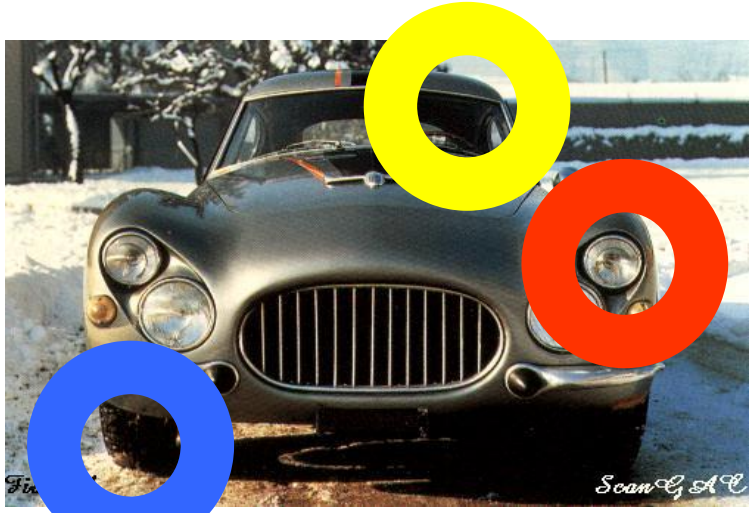
Figure from Winn & Shotton, CVPR '06

Generative part-based models









R. Fergus, P. Perona and A. Zisserman, Object Class Recognition by Unsupervised Scale-Invariant Learning, CVPR 2003

Probabilistic model

$$P(image \mid object) = P(appearance, shape \mid object)$$

Part descriptors

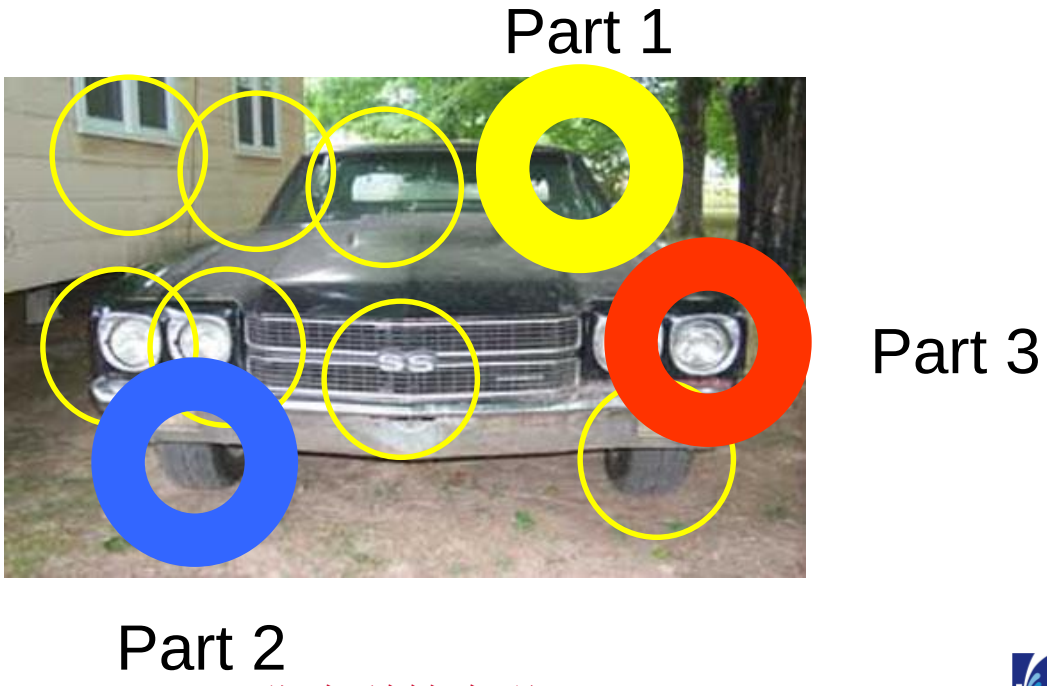
Part locations



Candidate parts

Probabilistic model

$$P(image \mid object) = P(appearance, shape \mid object)$$



Probabilistic model

$$P(image \mid object) = P(appearance, shape \mid object)$$

$$= \max_h P(appearance \mid h, object)\, p(shape \mid h, object)\, p(h \mid object)$$

h: assignment of features to parts

Part 1



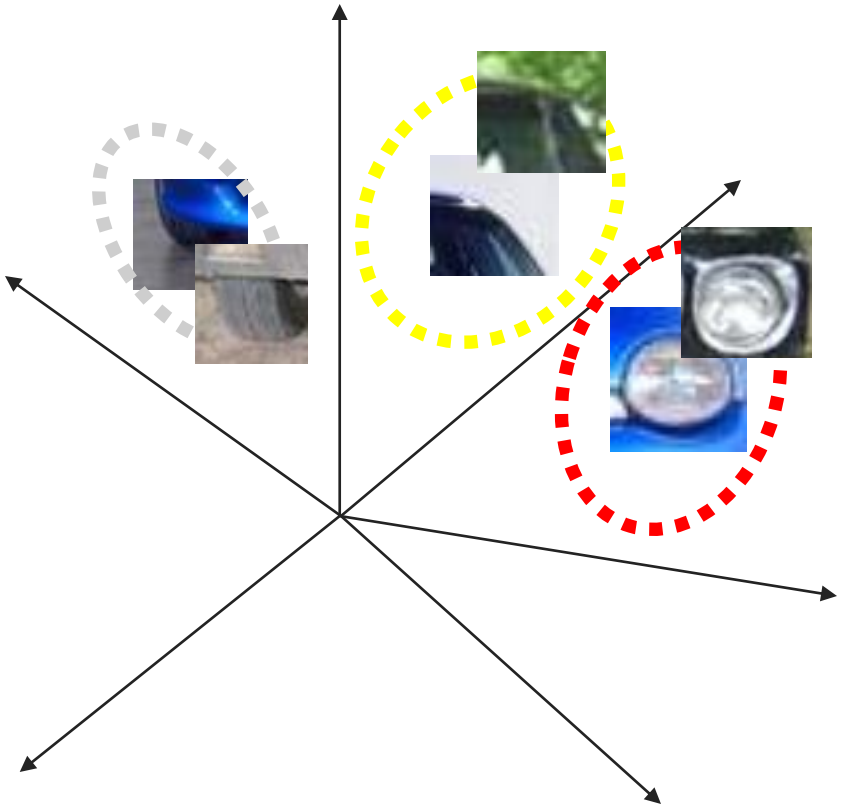
Part 3

Part 2

Probabilistic model

$$P(image \mid object) = P(appearance, shape \mid object)$$

$$= \max_h P(appearance \mid h, object)\, p(shape \mid h, object)\, p(h \mid object)$$



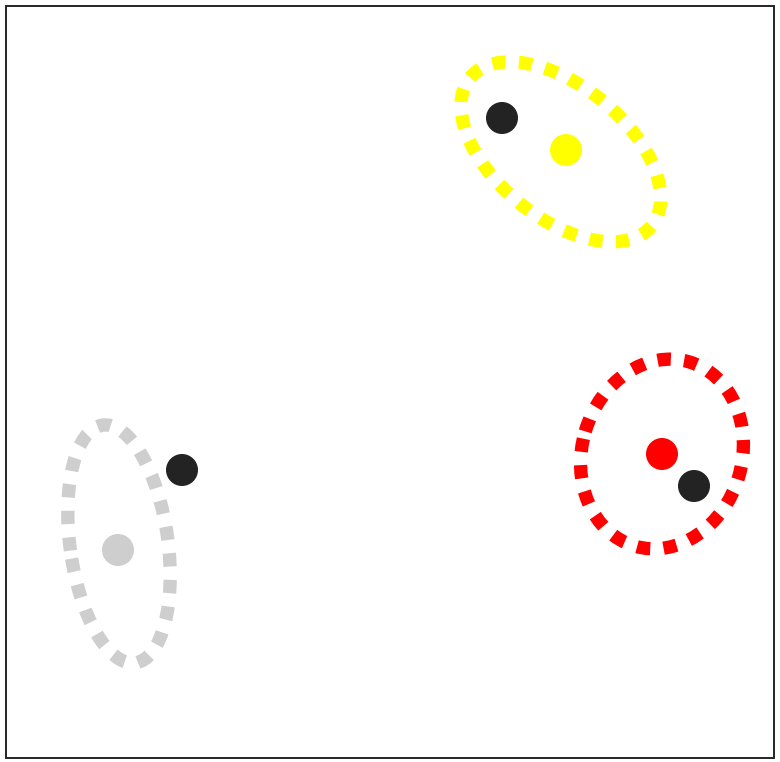
Distribution over patch descriptors

High-dimensional appearance space

Probabilistic model

$$P(image \mid object) = P(appearance, shape \mid object)$$

$$= \max_h P(appearance \mid h, object)\, p(shape \mid h, object)\, p(h \mid object)$$



Distribution over joint part positions

2D image space

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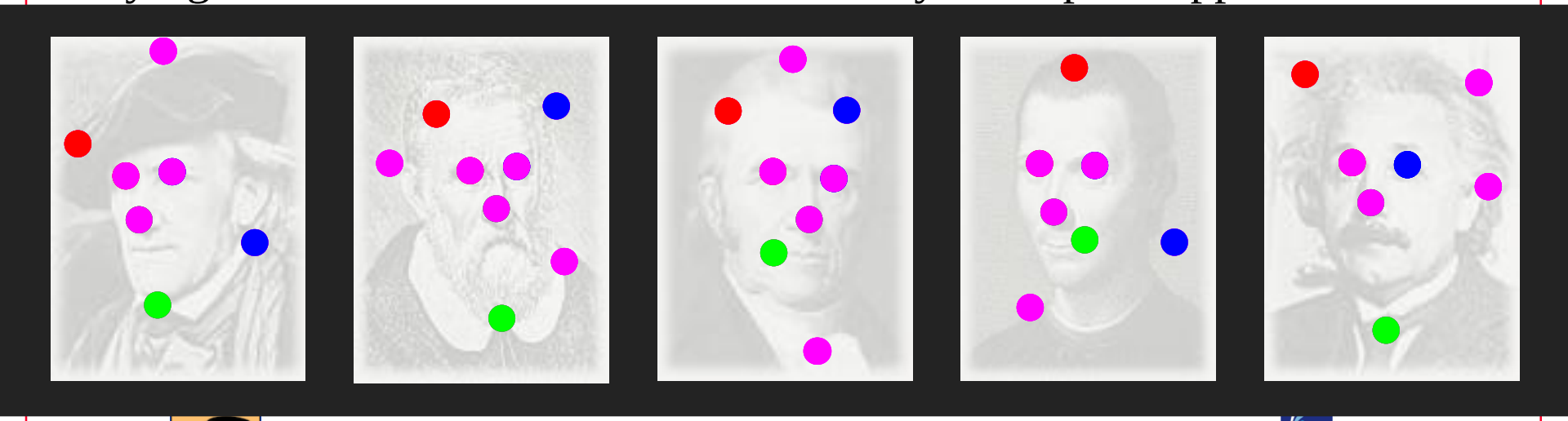
Learning procedure

- Find regions & their location & appearance
- Initialize model parameters
- Use EM and iterate to convergence:

  E-step: Compute assignments for which regions belong to which part

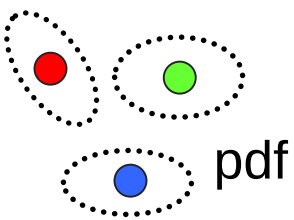
  M-step: Update model parameters

- Trying to maximize likelihood – consistency in shape & appearance

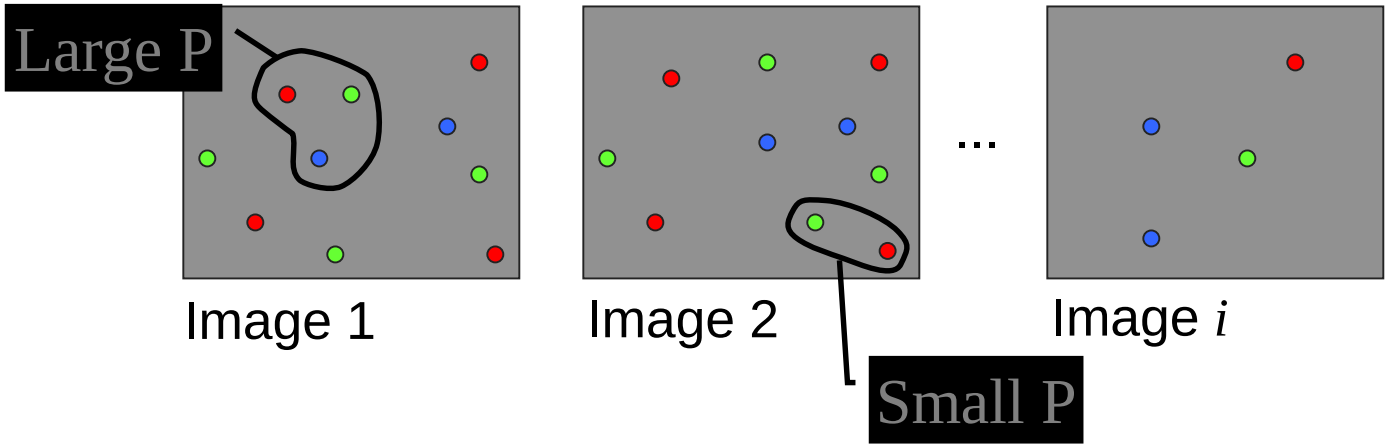


Example scheme, using EM for maximum likelihood learning

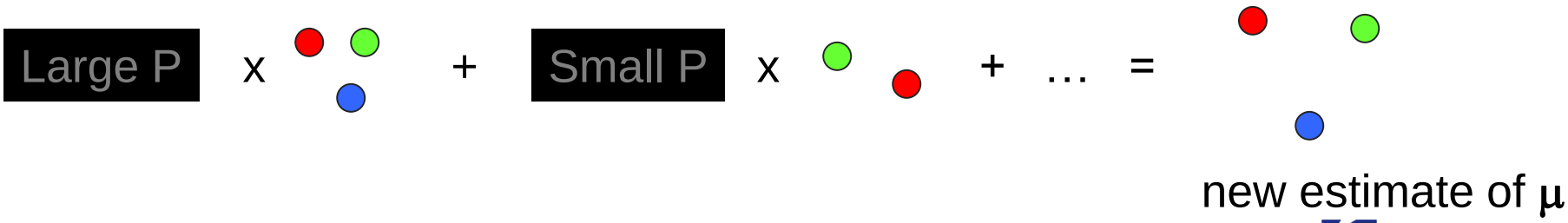
1. Current estimate of θ



2. Assign probabilities to constellations

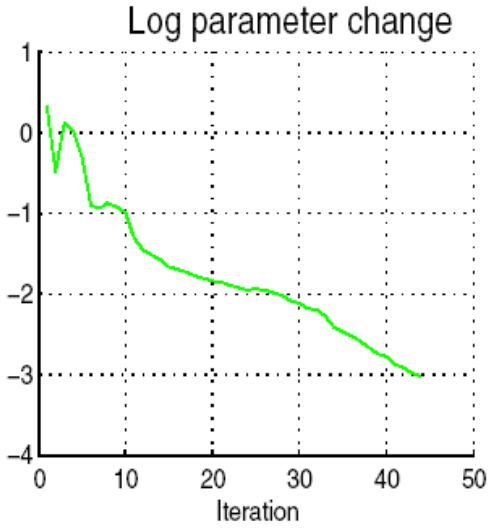


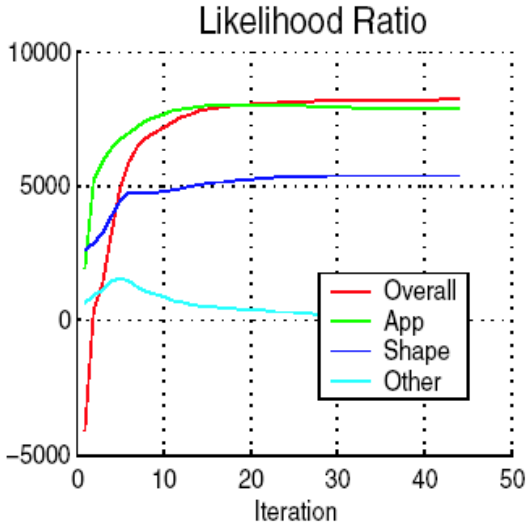
3. Use probabilities as weights to re-estimate parameters. Example: μ

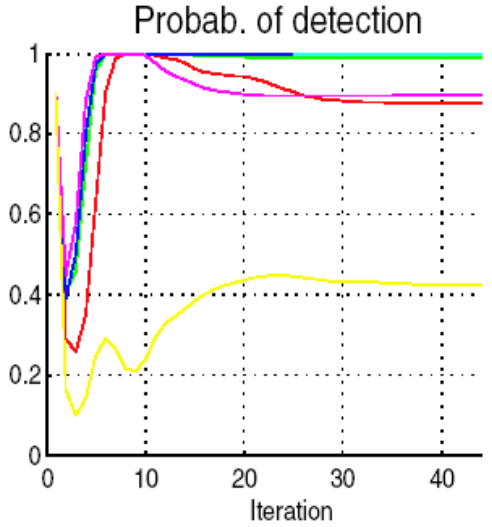


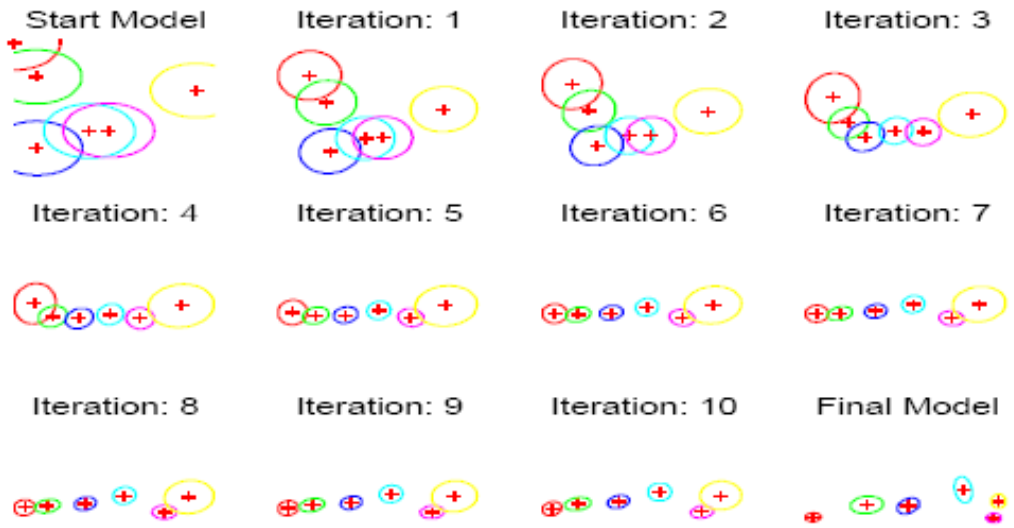
Learning Shape & Appearance simultaneously

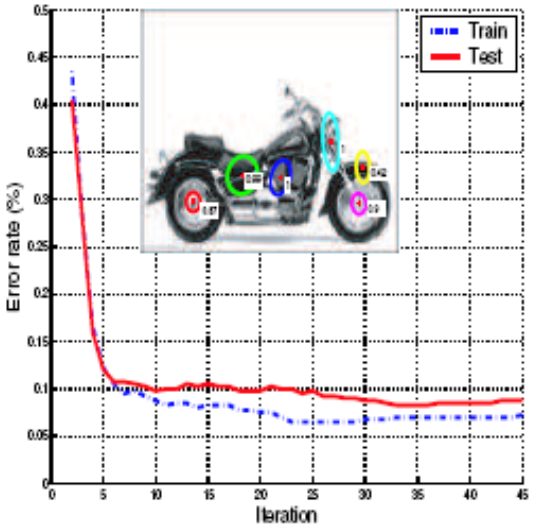
Fergus, Perona, Zisserman CVPR'03





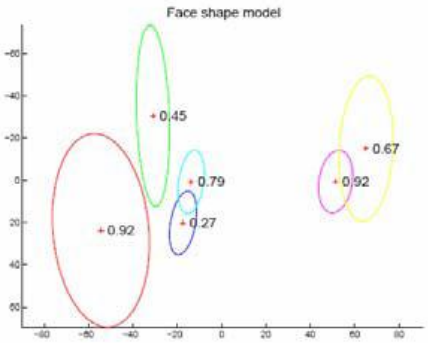






Results: Faces

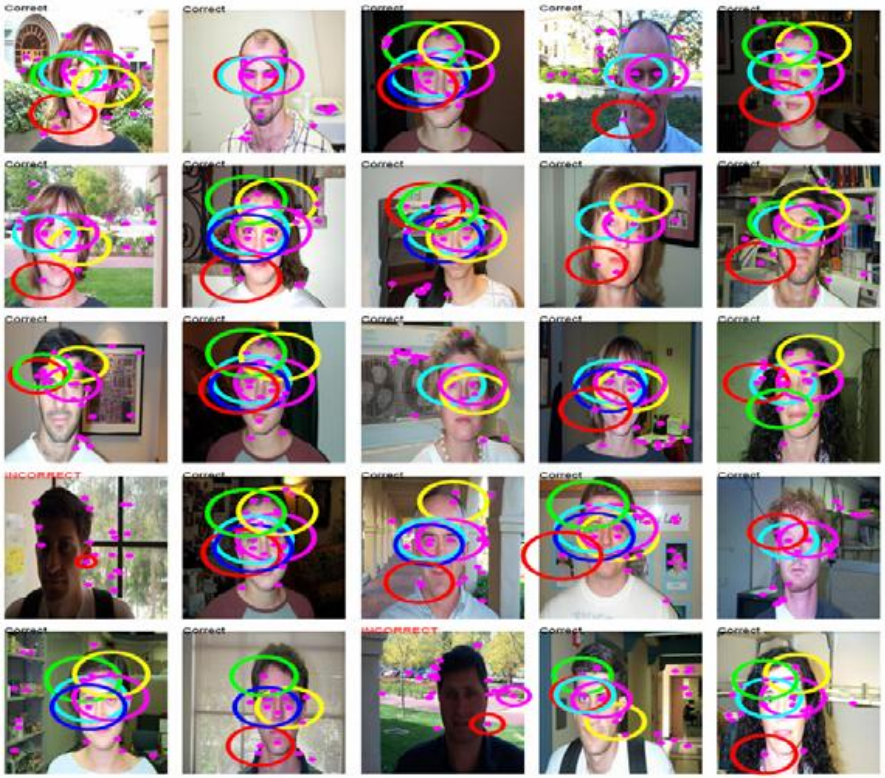
Face shape model



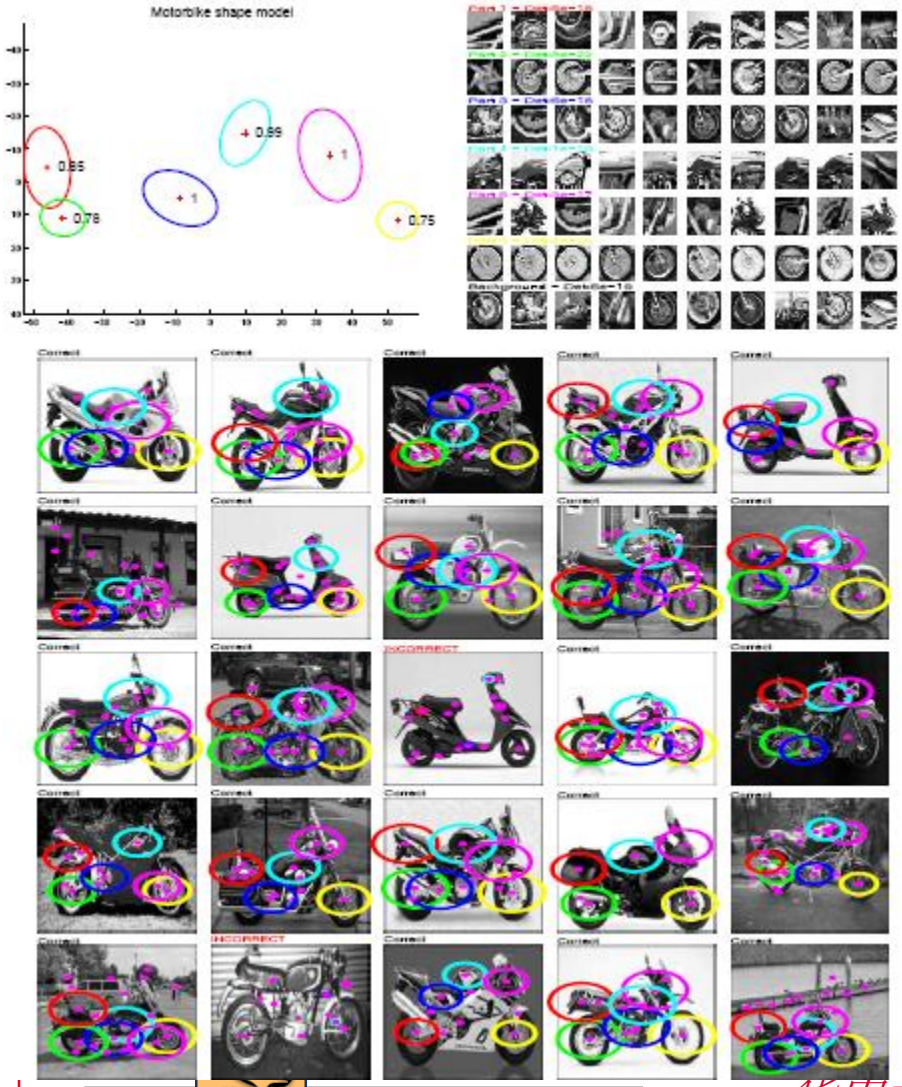


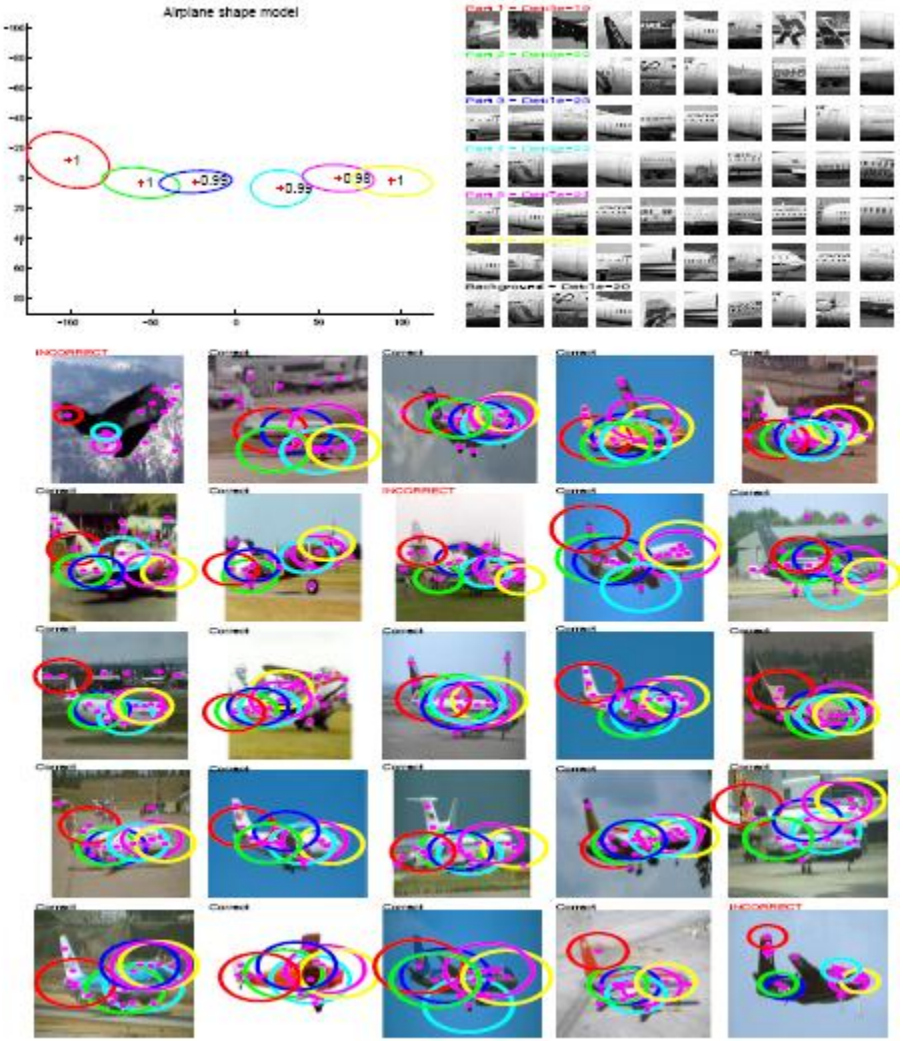
Patch appearance model

Recognition results



Results: Motorbikes and airplanes

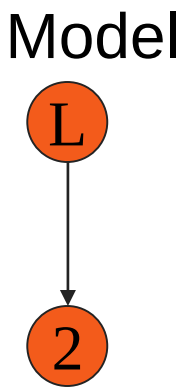




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Distance transforms

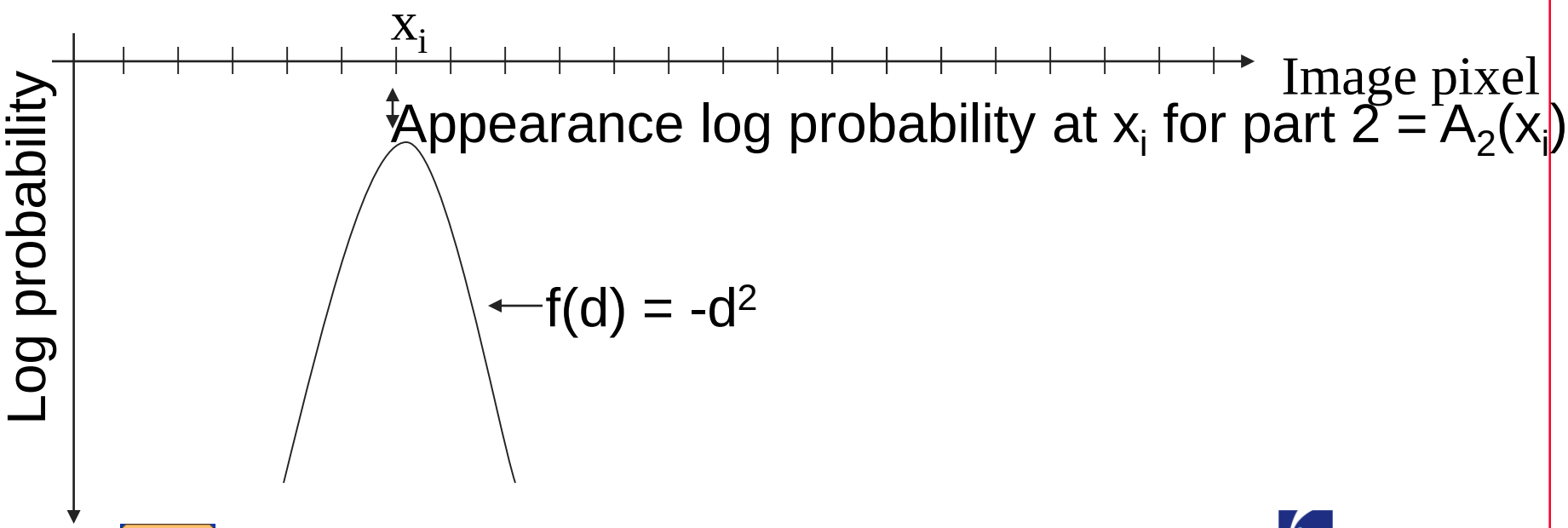


- Felzenszwalb and Huttenlocher CVPR'00 & IJCV'05
- Distance transforms

  $O(N^2P) \rightarrow O(NP)$ for tree structured models (1-fan)
- How it works

  Assume location model is Gaussian (i.e. e^{-d^2})

  Consider a two part model with $\mu=0$, $\sigma=1$ on a 1-D image



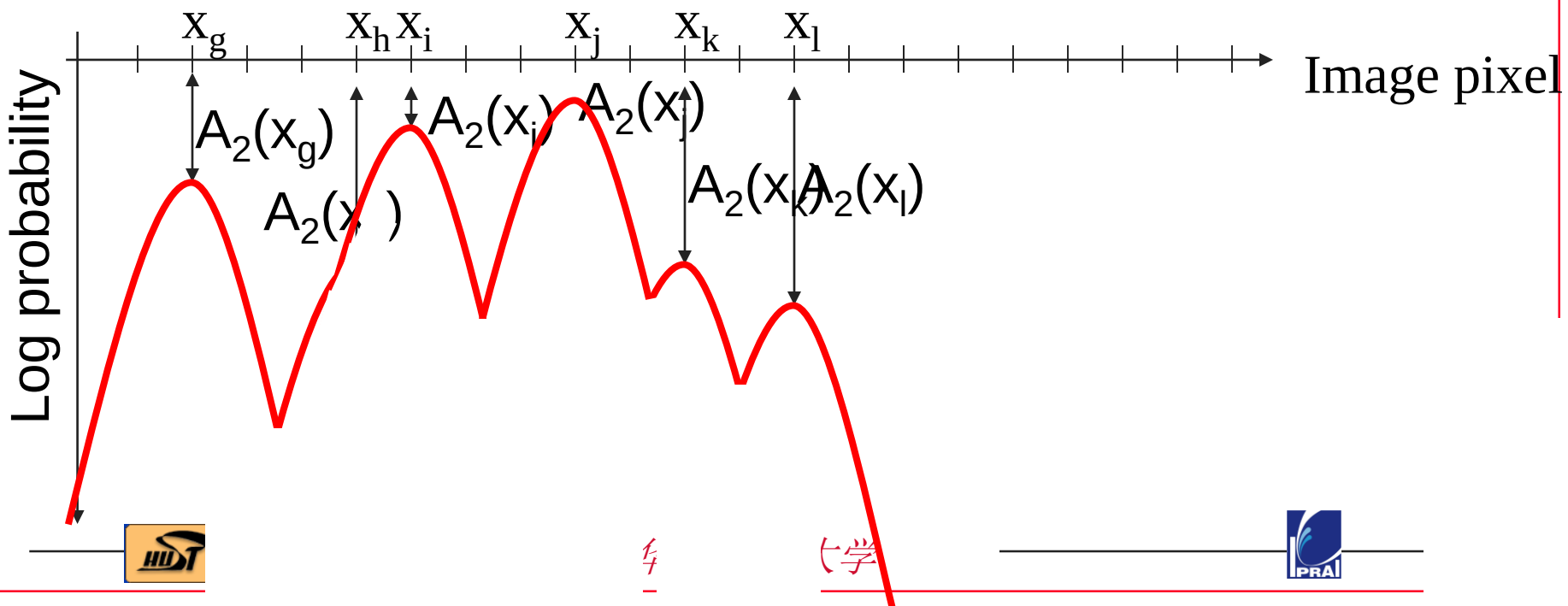
Distance transforms 2

- For each position of landmark part, find best position for part 2

  Finding most probable x_i is equivalent finding maximum over set of offset parabolas

  Upper envelope computed in O(N) rather than obvious $O(N^2)$ via distance transform (see Felzenszwalb and Huttenlocher IJCV'05).

- Add $A_L(x)$ to upper envelope (offset by μ) to get overall probability map

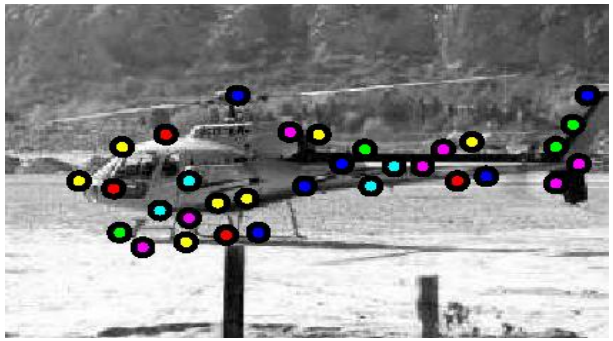


Overview

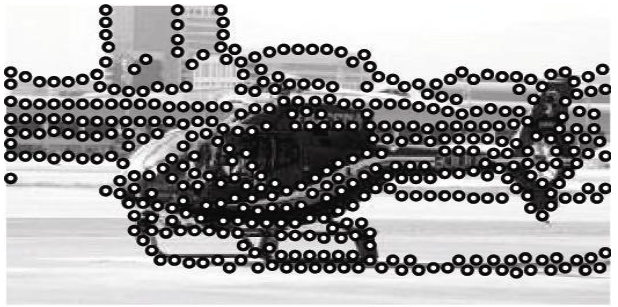
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Deformable Template Matching

Berg et al. CVPR 2005



Template



Query

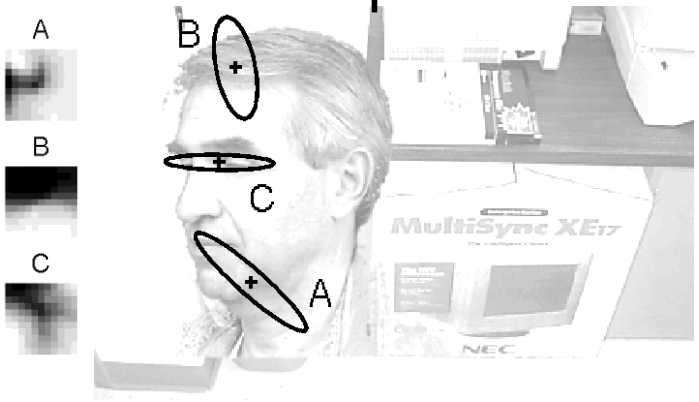
- Formulate problem as Integer Quadratic Programming
- $O(N^P)$ in general
- Use approximations that allow P=50 and N=2550 in <2 secs



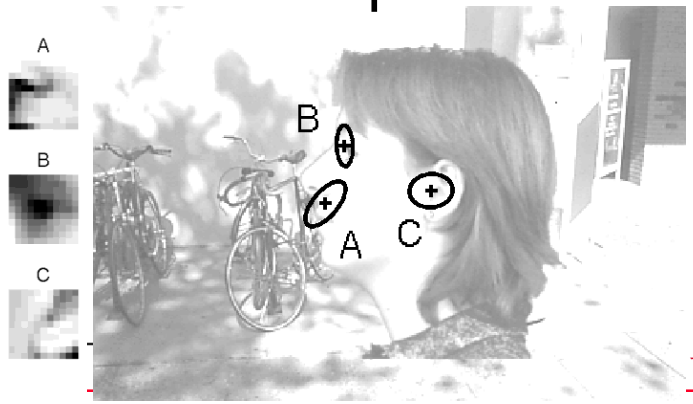
Multiple views

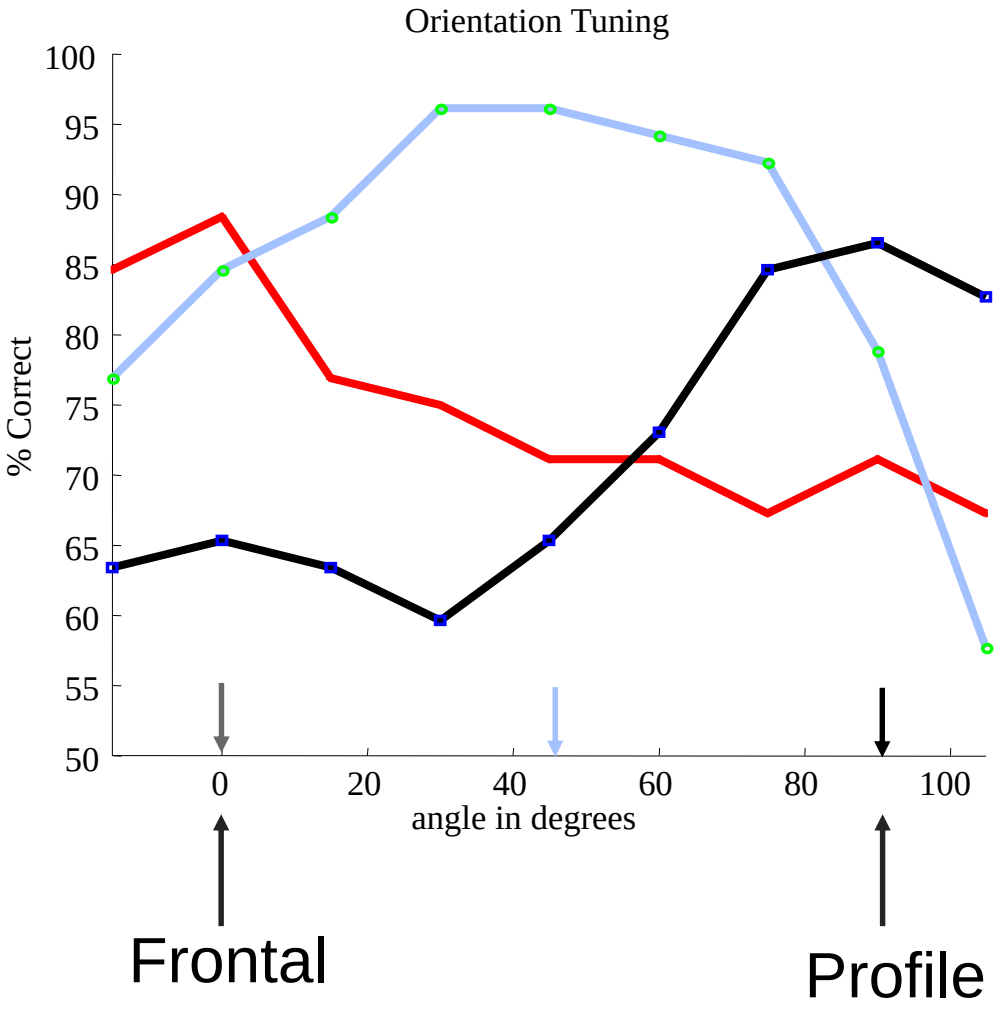
- Full 3-D location model
- Mixture of 2-D models
  - Weber CVPR ‘00

Component 1



Component 2



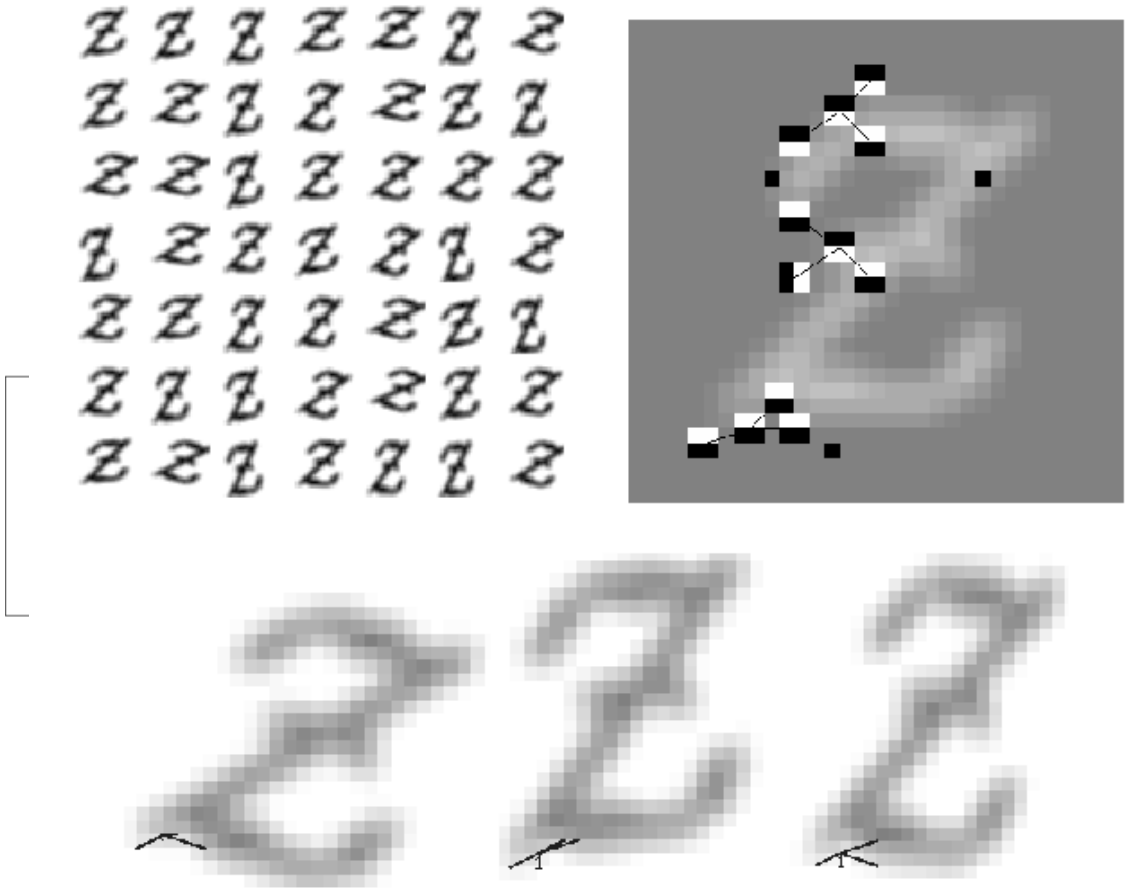


Hierarchical Representations

- Pixels → Pixel groupings → Parts → Object

- Multi-scale approach increases number of low-level features

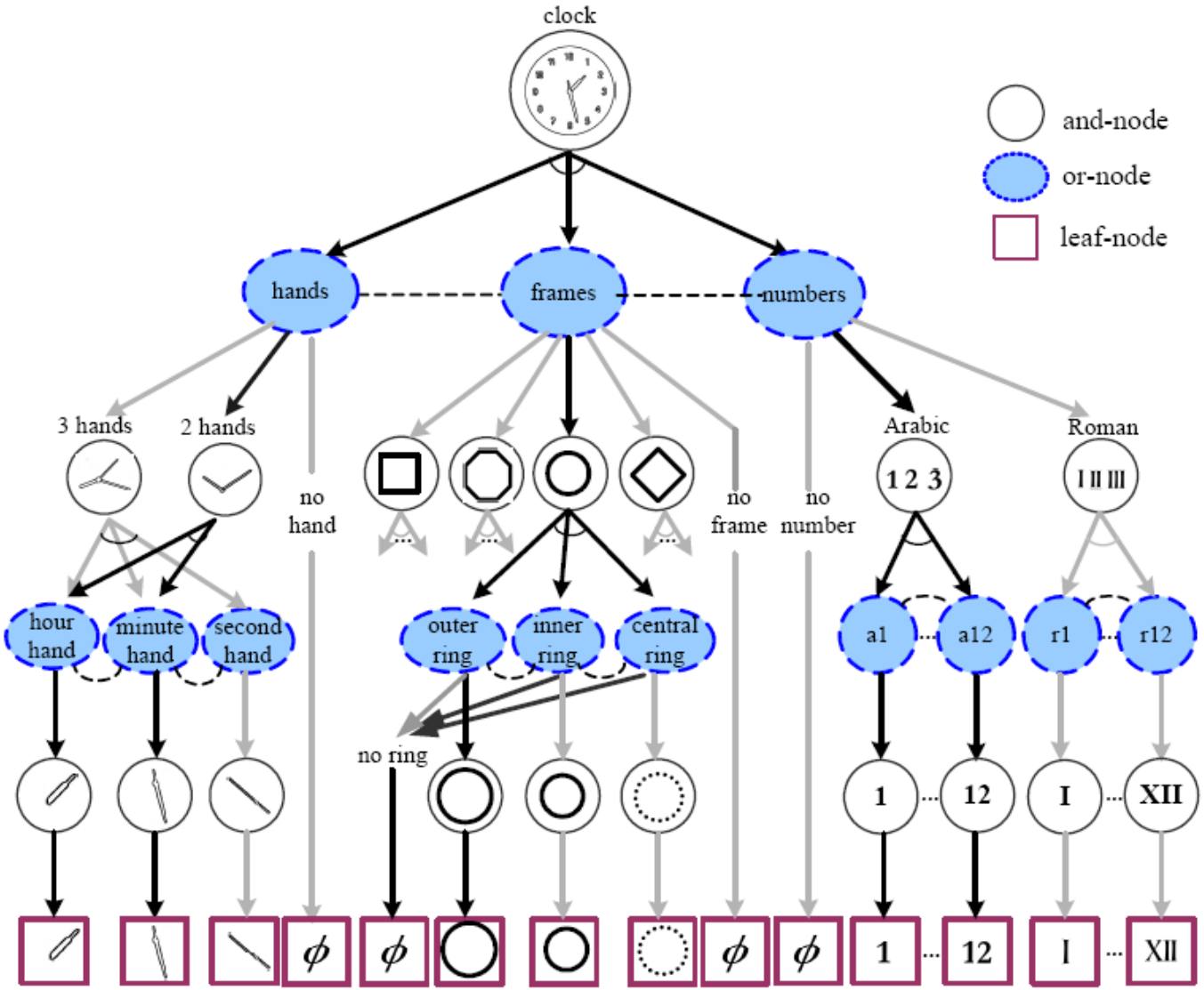
- Amit and Geman ’98
- Ullman et al.
- Bouchard & Triggs ’05
- Zhu and Mumford
- Jin & Geman ‘06
- Zhu & Yuille ’07
- Fidler & Leonardis ’07



Images from [Amit98]

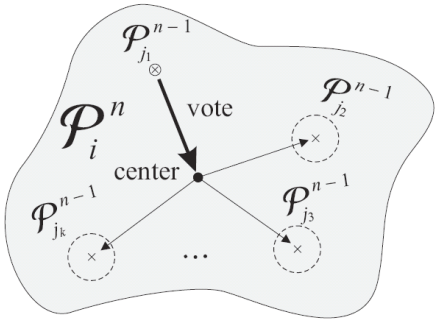
Stochastic Grammar of Images

S.C. Zhu et al. and D. Mumford

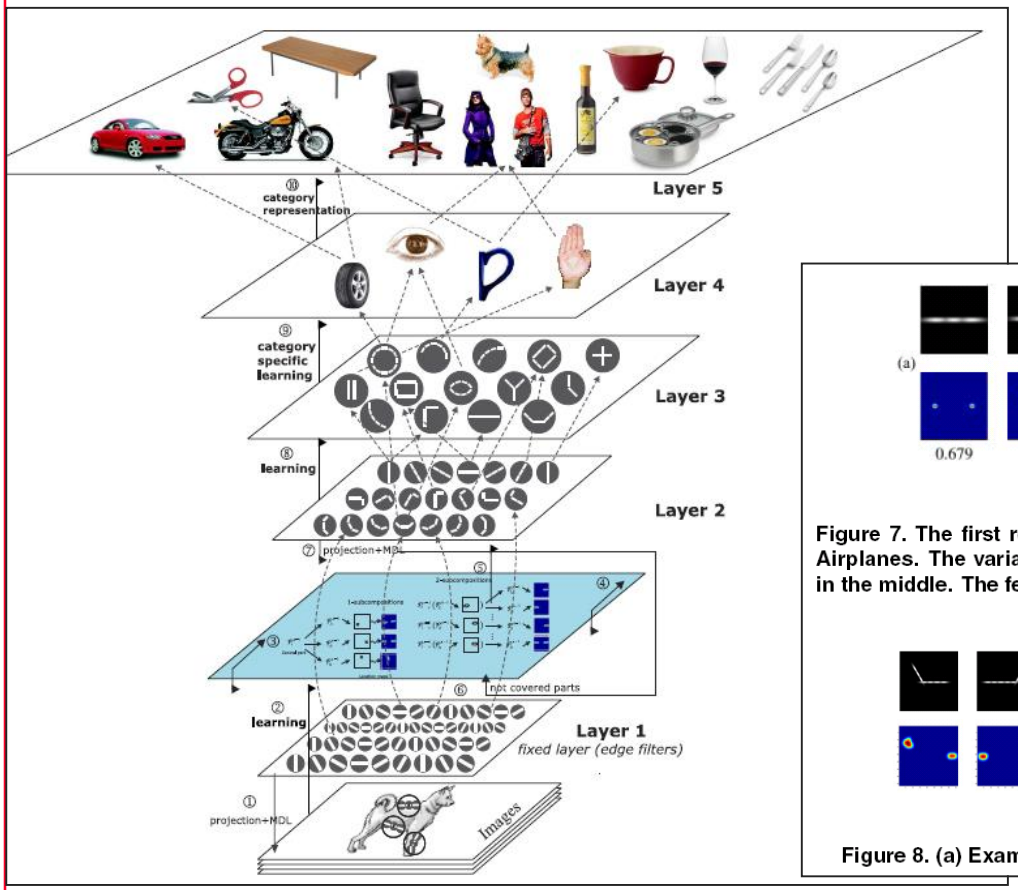


Learning a Compositional Hierarchy of Object Structure

Fidler & Leonardis, CVPR'07; Fidler, Boben & Leonardis, CVPR 2008



Parts model



The architecture

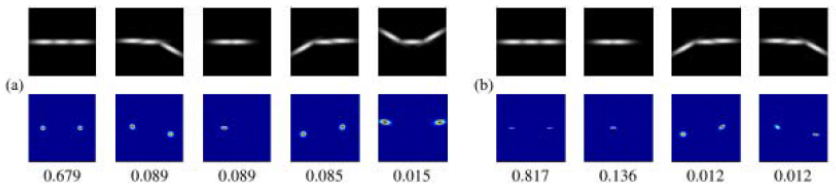


Figure 7. The first row depicts the final parts comprising Layer II obtained for (a) Cliparts and (b) Airplanes. The variances of position distributions of parts, relative to the central part, are depicted in the middle. The feature probabilities are listed in the last row.

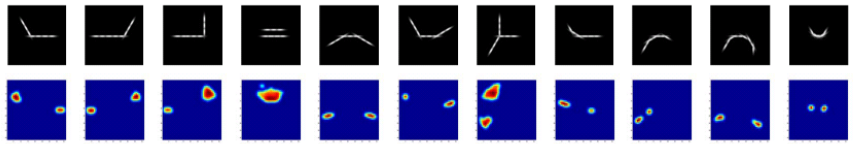
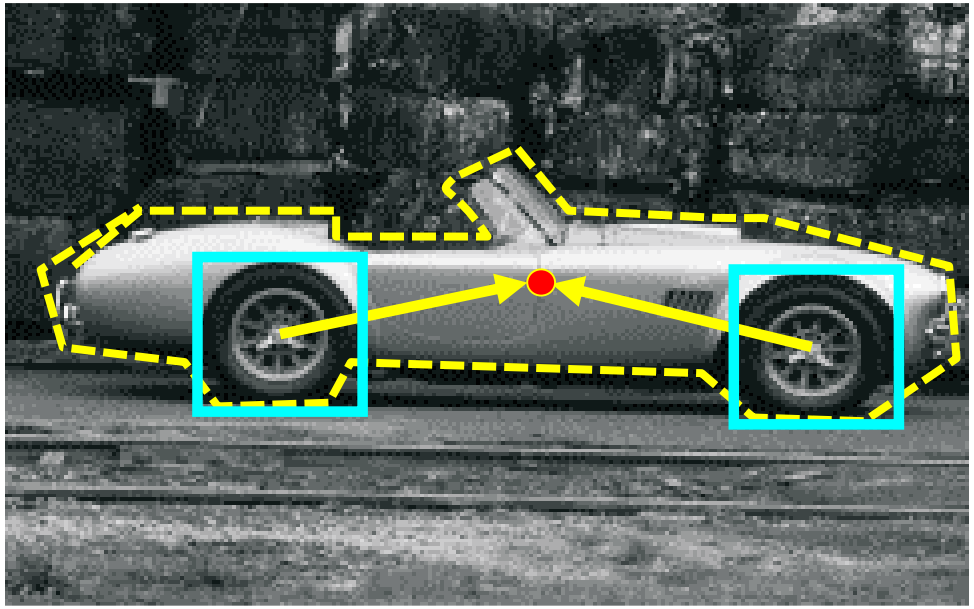


Figure 8. (a) Examples of Layer 3 parts, (b) variances of positions of the surrounding subparts

Learned parts

Implicit shape models

- Visual codebook is used to index votes for object position





visual codeword with displacement vectors

training image annotated with object localization info

B. Leibe, A. Leonardis, and B. Schiele, Combined Object Categorization and Segmentation with an Implicit Shape Model, ECCV Workshop on Statistical Learning in Computer Vision 2004

Implicit shape models

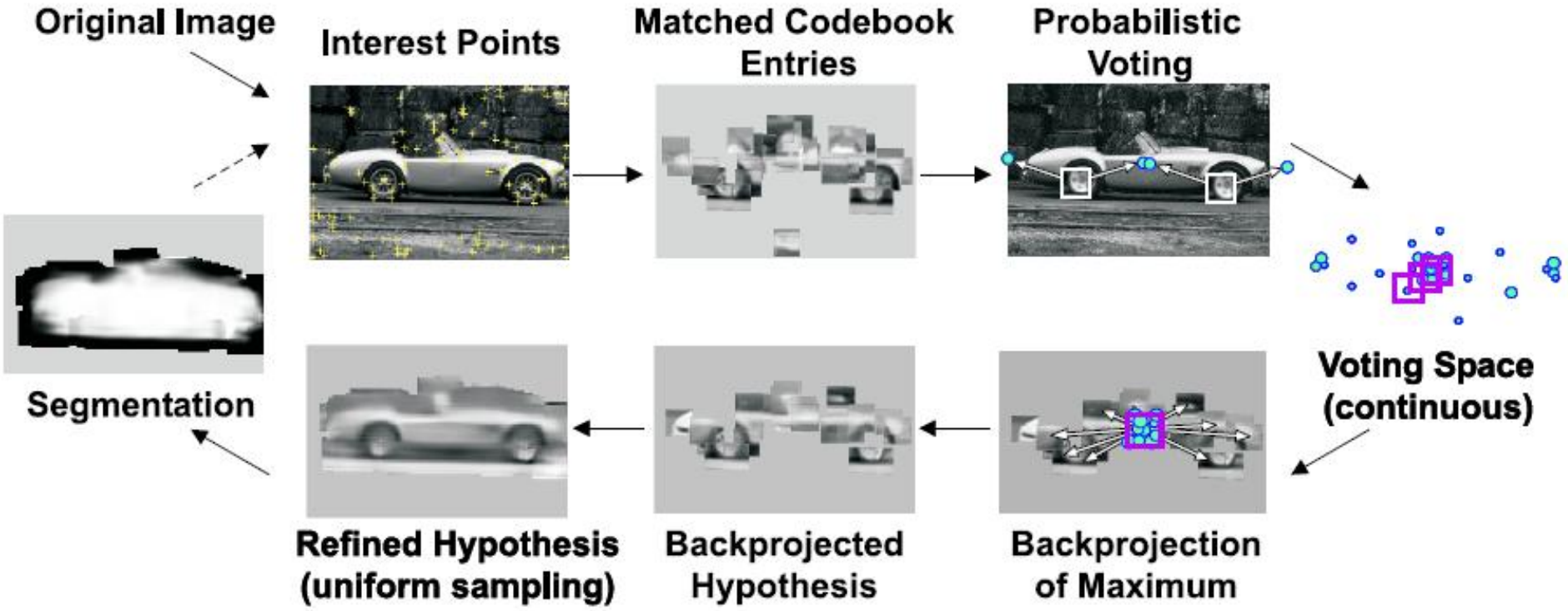
- Visual codebook is used to index votes for object position



test image

B. Leibe, A. Leonardis, and B. Schiele, Combined Object Categorization and Segmentation with an Implicit Shape Model, ECCV Workshop on Statistical Learning in Computer Vision 2004

Implicit shape models: Details



B. Leibe, A. Leonardis, and B. Schiele, Combined Object Categorization and Segmentation with an Implicit Shape Model, ECCV Workshop on Statistical Learning in Computer Vision 2004

Overview

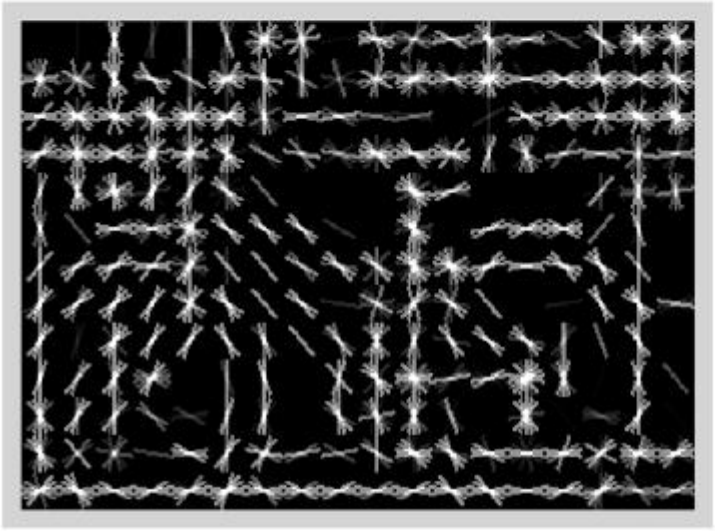
- Representation
  - Location
  - Appearance
  - Generative interpretation
- Learning
- Distance transforms
- Other approaches using parts
- Felzenszwalb, McAllester, Ramanan CVPR2008 / Felzenszwalb, Girshick, McAllester, Ramanan TPAMI2010

Object Detection with Discriminatively Trained Part Based Models

Pedro F. Felzenszwalb, Ross B. Girshick, David McAllester and Deva Ramanan

Histogram of Gradient (HOG) features

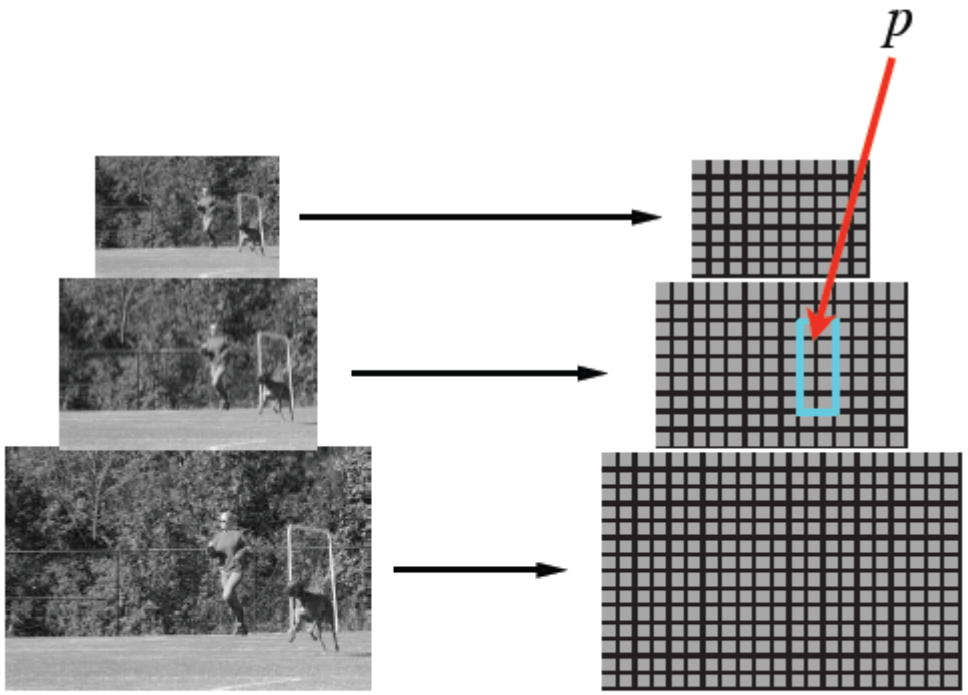




- Image is partitioned into 8x8 pixel blocks
- In each block we compute a histogram of gradient orientations
  - Invariant to changes in lighting, small deformations, etc.
- Compute features at different resolutions (pyramid)

HOG Filters

- Array of weights for features in subwindow of HOG pyramid
- Score is dot product of filter and feature vector



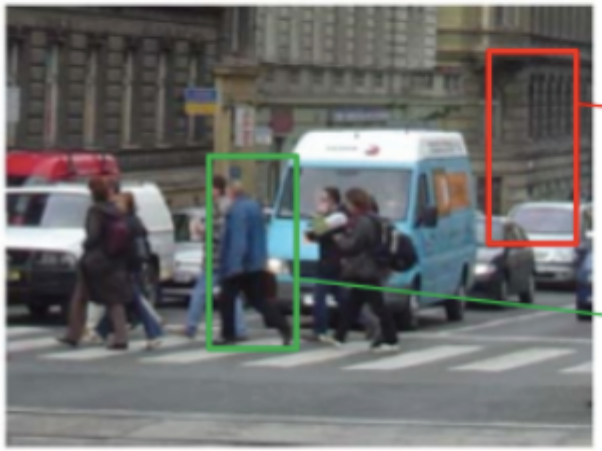
Filter F

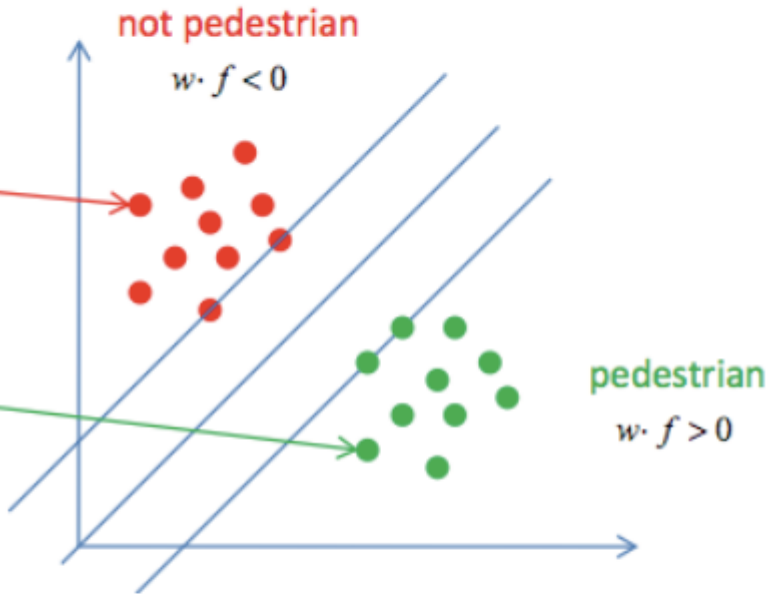


Score of F at position p is $F \cdot \phi(p, H)$

$\phi(p, H)$ = concatenation of HOG features from subwindow specified by p

Dalal & Triggs: HOG + linear SVMs







Typical form of a model

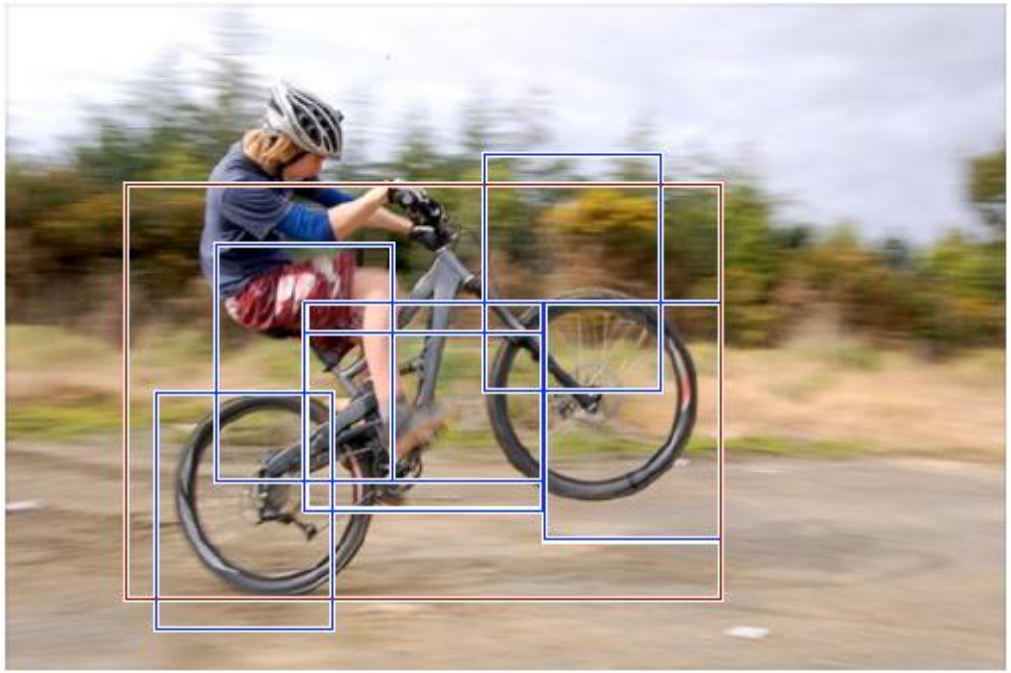
There is much more background than objects

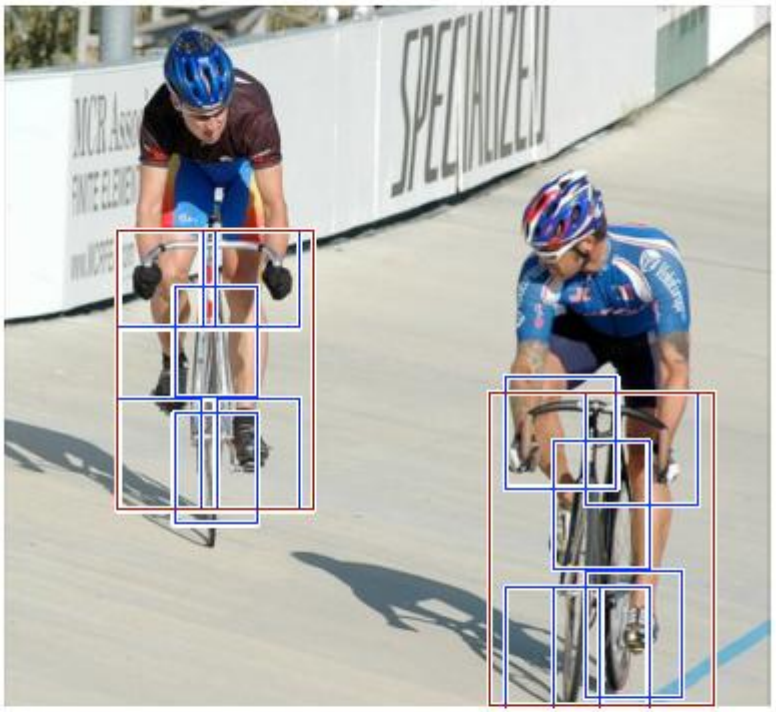
Start with random negatives and repeat:

1) Train a model

2) Harvest false positives to define "hard negatives"

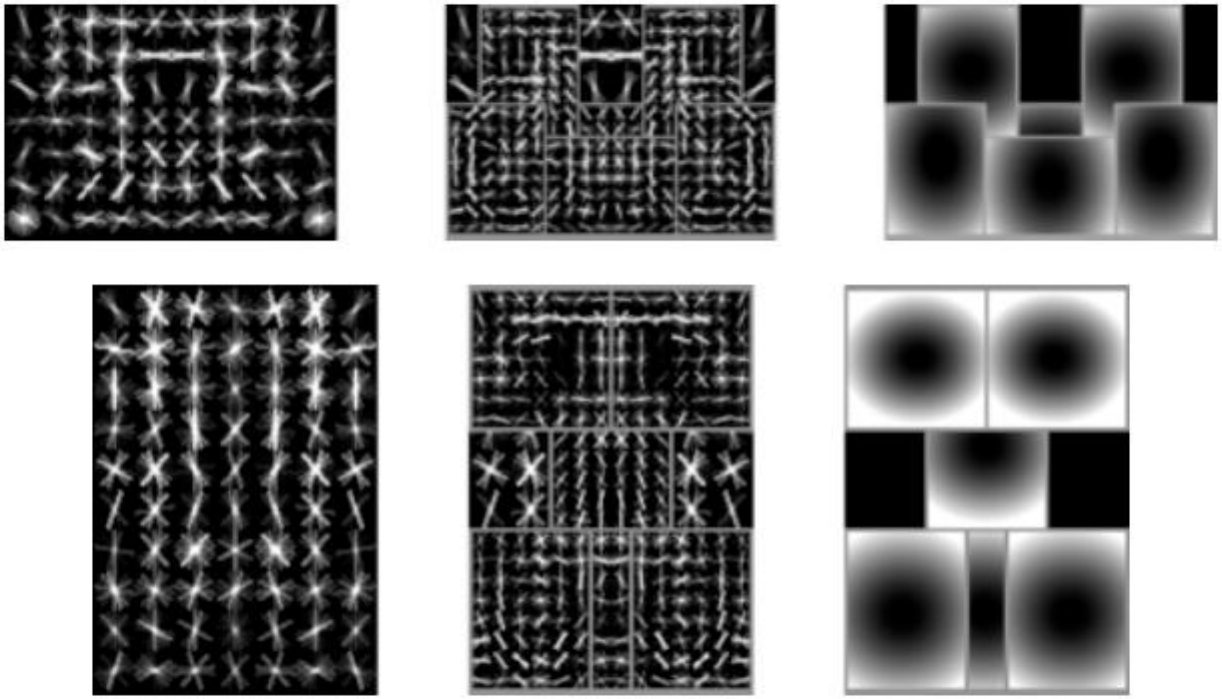
Overview of our models





- Mixture of deformable part models
- Each component has global template + deformable parts
- Fully trained from bounding boxes alone

2 component bicycle model



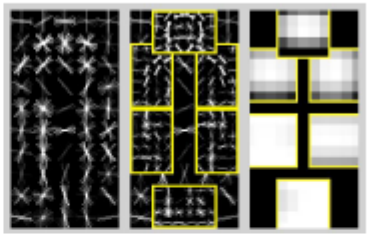
root filters
coarse resolution

part filters
finer resolution

deformation
models

Each component has a root filter F_0
and n part models (F_i, v_i, d_i)

Object hypothesis



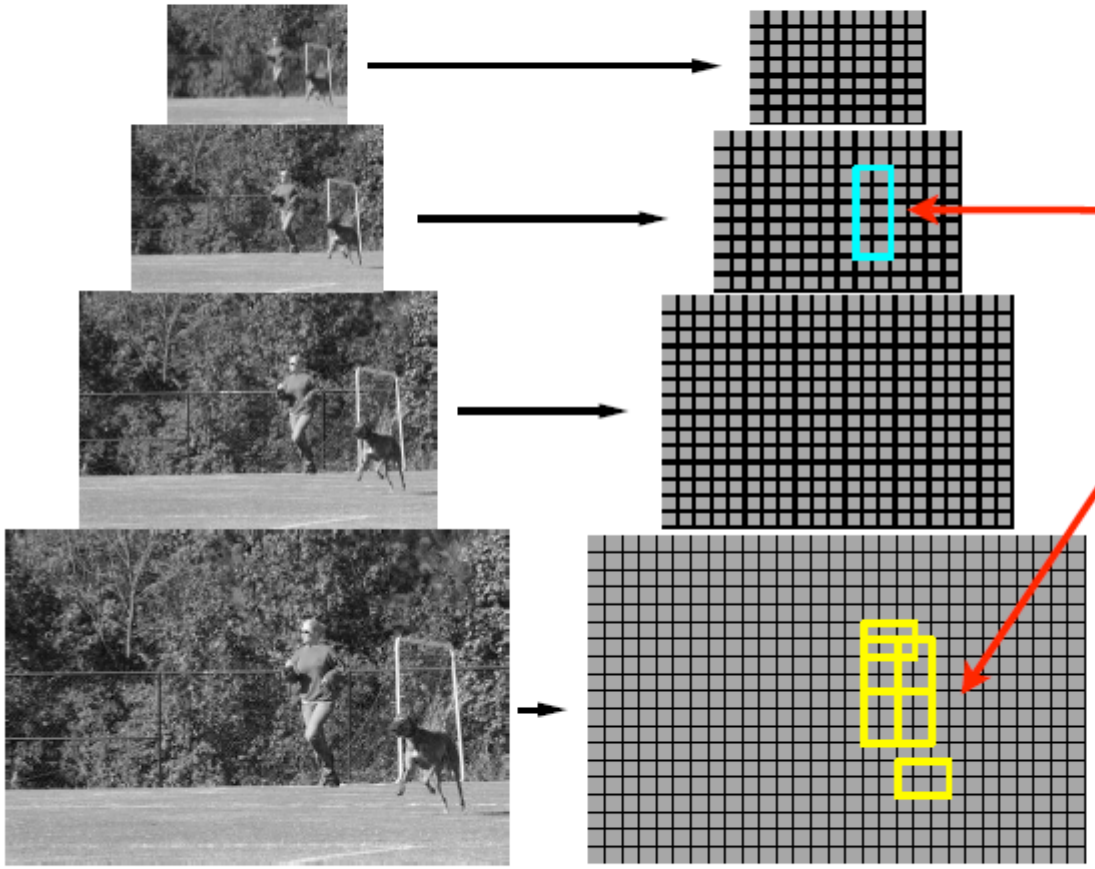


Image pyramid

HOG feature pyramid

$z = (p_0, \ldots, p_n)$

p_0 : location of root

$p_1, \ldots, p_n$: location of parts

Score is sum of filter scores minus deformation costs

Multiscale model captures features at two-resolutions

Score of a hypothesis

$$\text{score}(p_0, \ldots, p_n) = \underbrace{\sum_{i=0}^{n} F_i \cdot \phi(H, p_i)}_{\text{“data term”}} - \underbrace{\sum_{i=1}^{n} d_i \cdot (dx_i^2, dy_i^2)}_{\text{“spatial prior”}}$$

F_i: filters

d_i: displacements

(dx_i^2, dy_i^2): deformation parameters



$$\text{score}(z) = \beta \cdot \Psi(H, z)$$

β: concatenation filters and deformation parameters

$\Psi(H, z)$: concatenation of HOG features and part displacement features

Matching

- Define an overall score for each root location
  - Based on best placement of parts

$$\text{score}(p_0) = \max_{p_1,\ldots,p_n} \text{score}(p_0,\ldots,p_n).$$

- High scoring root locations define detections
  - "sliding window approach"
- Efficient computation: dynamic programming + generalized distance transforms (max-convolution)



head filter

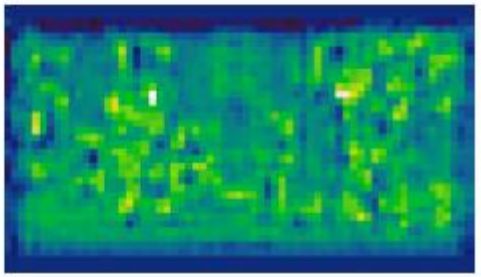
input image



Response of filter in l-th pyramid level

$$R_l(x, y) = F \cdot \phi(H, (x, y, l))$$

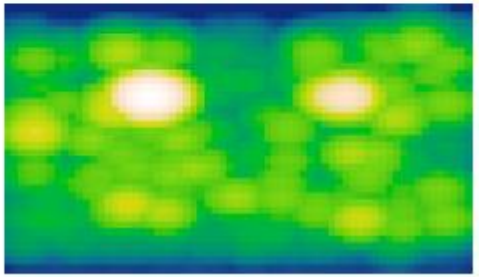
cross-correlation

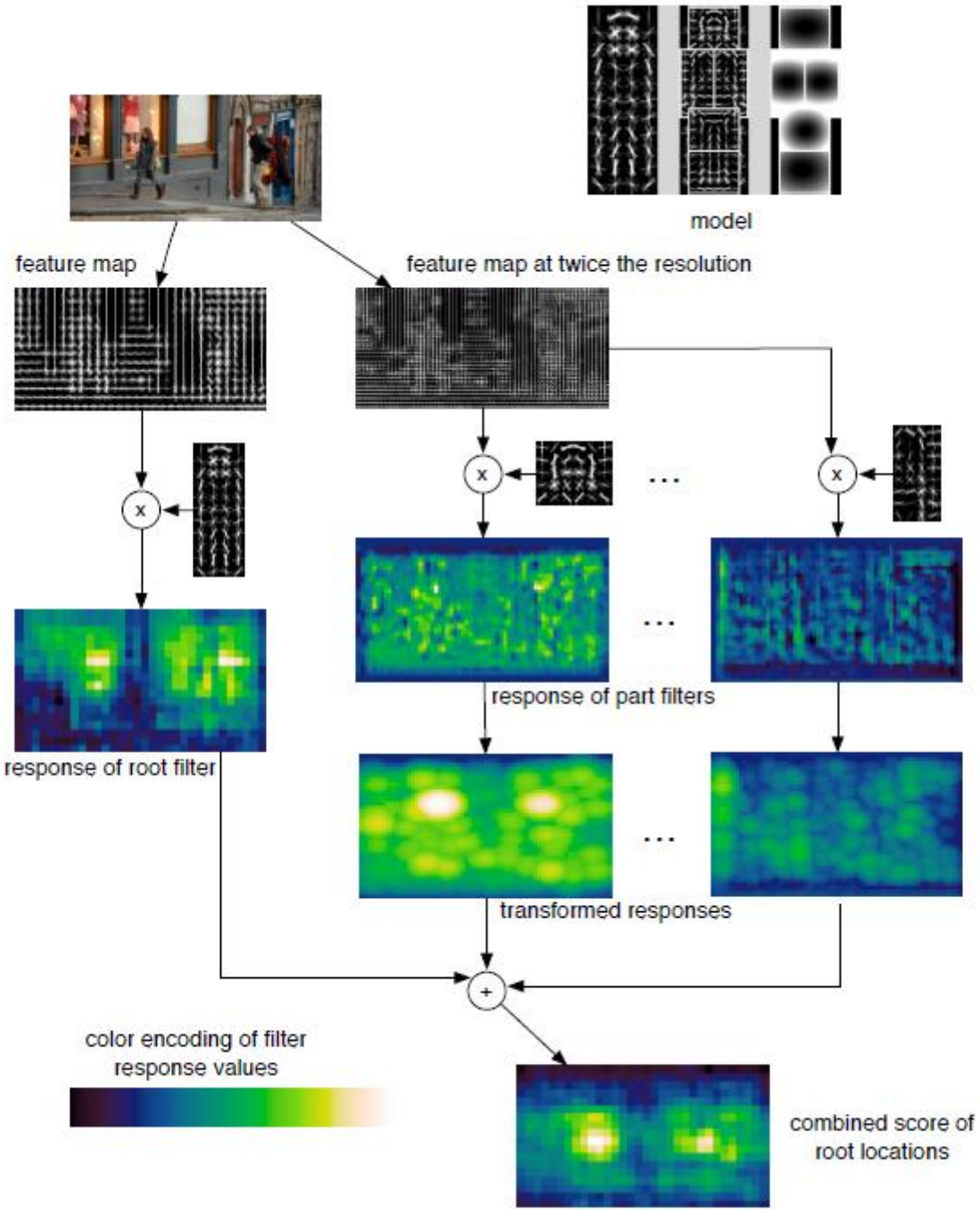


Transformed response

$$D_l(x, y) = \max_{dx,dy} \left(R_l(x + dx, y + dy) - d_i \cdot (dx^2, dy^2) \right)$$

max-convolution, computed in linear time
(spreading, local max, etc)



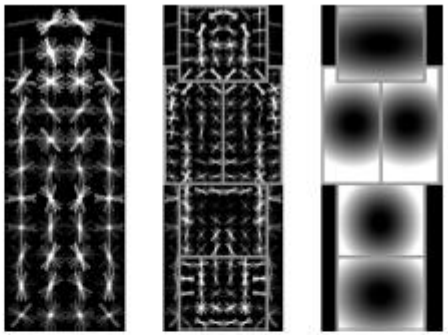

model
feature map
feature map at twice the resolution
response of part filters
response of root filter
transformed responses
color encoding of filter
response values
combined score of
root locations

Training

- Training data consists of images with labeled bounding boxes.
- Need to learn the model structure, filters and deformation costs.



Training



Latent SVM (MI-SVM)

Classifiers that score an example x using

$$f_\beta(x) = \max_{z \in Z(x)} \beta \cdot \Phi(x, z)$$

β are model parameters

z are latent values

正样本的root偏移z也是需要学习得到的

Training data $D = (\langle x_1, y_1 \rangle, \ldots, \langle x_n, y_n \rangle)$ $y_i \in \{-1, 1\}$

We would like to find β such that: $y_i f_\beta(x_i) > 0$

Minimize

$$L_D(\beta) = \frac{1}{2}||\beta||^2 + C \sum_{i=1}^{n} \max(0, 1 - y_i f_\beta(x_i))$$

Latent SVM training

$$L_D(\beta) = \frac{1}{2}||\beta||^2 + C\sum_{i=1}^{n}\max(0, 1 - y_i f_\beta(x_i))$$

- Convex if we fix z for positive examples
- Optimization:
  - Initialize β and iterate:
    - Pick best z for each positive example
    - Optimize β via gradient descent with data-mining

先用GT中的正样本训练得到初始root filter，再找出root filter响应最大、且与GT的overlap足够大的偏移z处样本作为正样本，重新训练root filter。然后固定正样本，挖掘难例负样本进行训练root filter。

Training algorithm, nested iterations

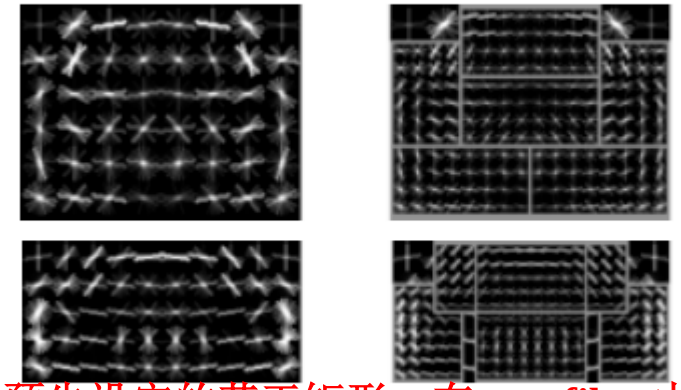
Fix "best" positive latent values for positives

- Harvest high scoring (x,z) pairs from background images
- Update model using gradient descent
- Trow away (x,z) pairs with low score

上：对应于前一slide的最后一步，即optimize beta

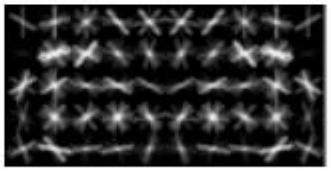
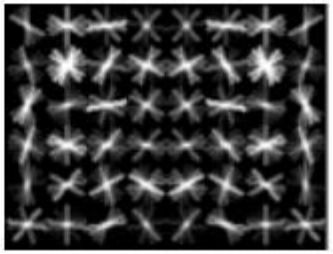
下：完整的步骤，其中的train对应于前一slide的整个optimization过程

- Sequence of training rounds
  - Train root filters
  - Initialize parts from root
  - Train final model

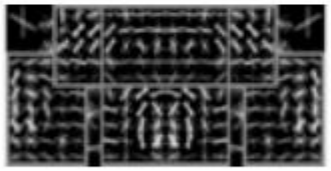
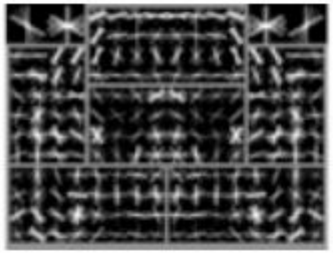


Part filter的初始化依据root filter的weight，根据预先设定的若干矩形，在root filter中寻找energy最大的矩形，作为当前确定的part形状及大小。将root filter中对应区域的weight置为0，继续寻找下一个energy最大的矩形，直到找到预定的part数（论文中为6）

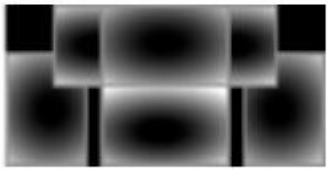
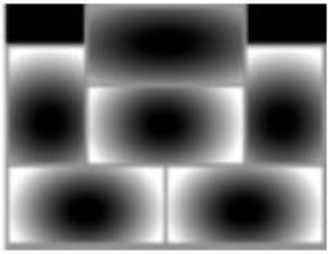
Car model

root filters
coarse resolution

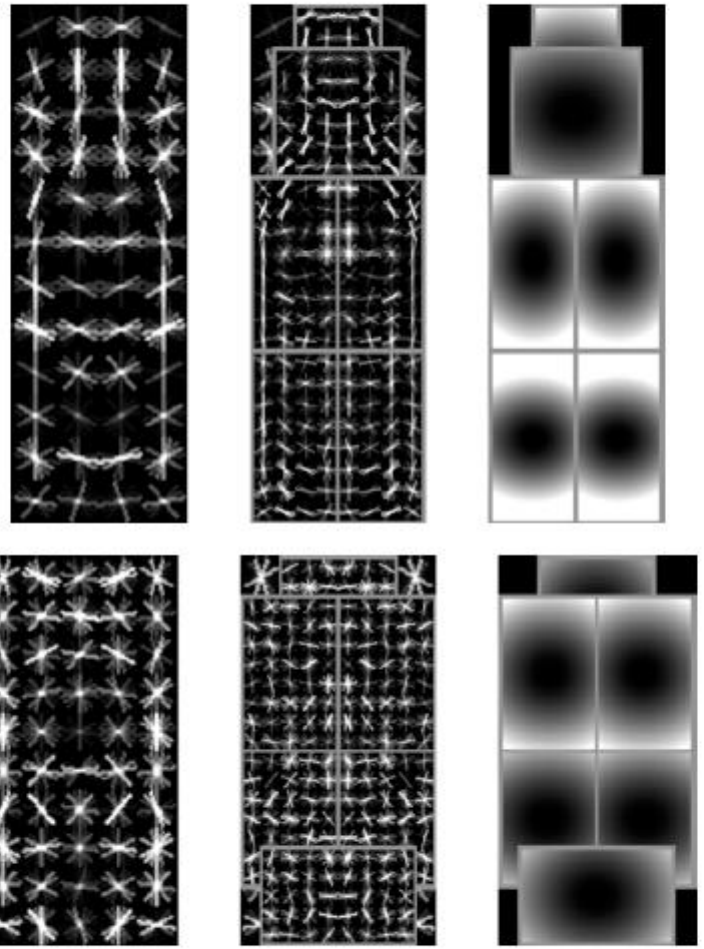



part filters
finer resolution

deformation
models

Bottle model



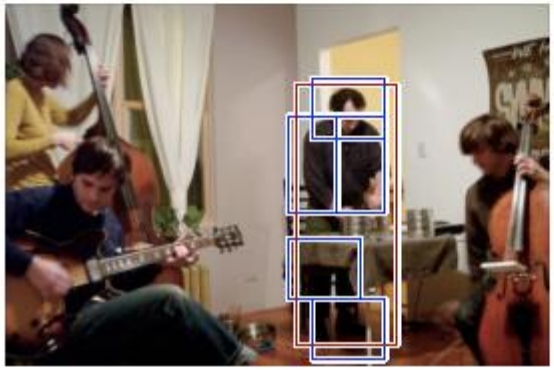
root filters
coarse resolution

part filters
finer resolution

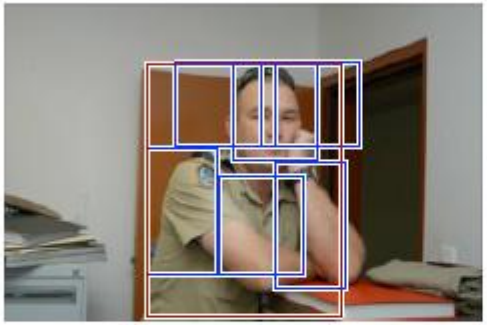
deformation
models

Person detections

high scoring true positives









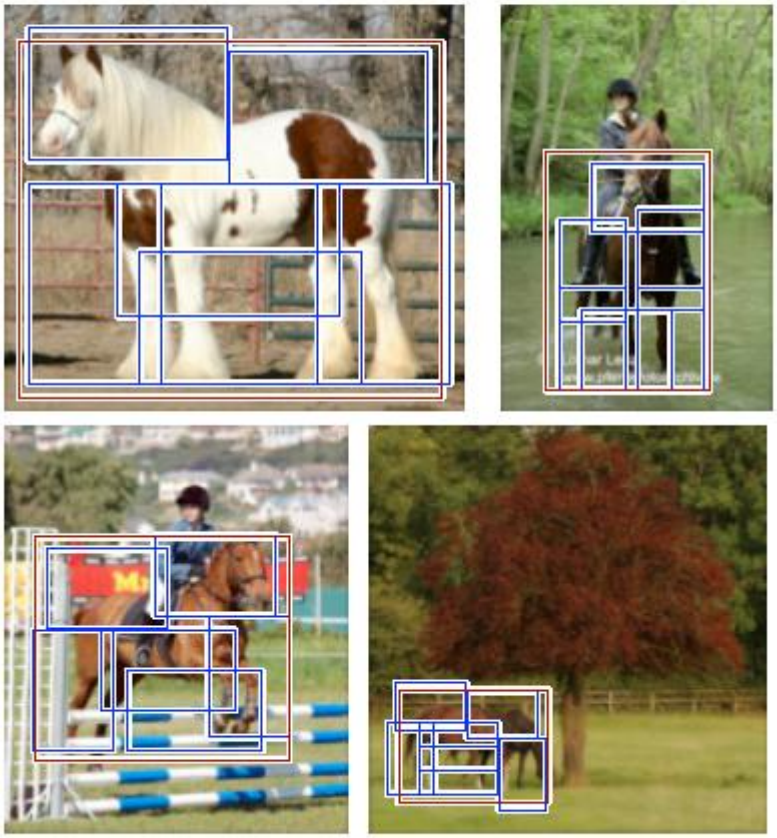
high scoring false positives
(not enough overlap)



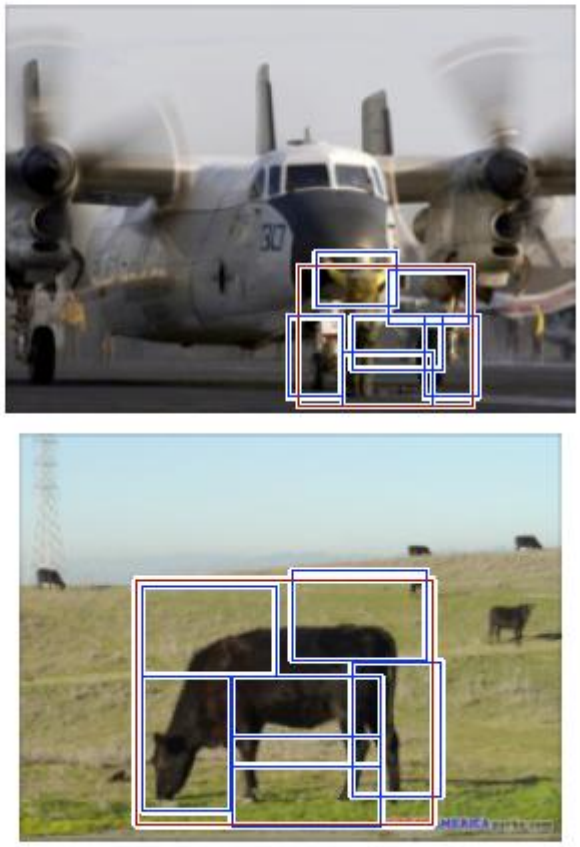


Horse detections

high scoring true positives

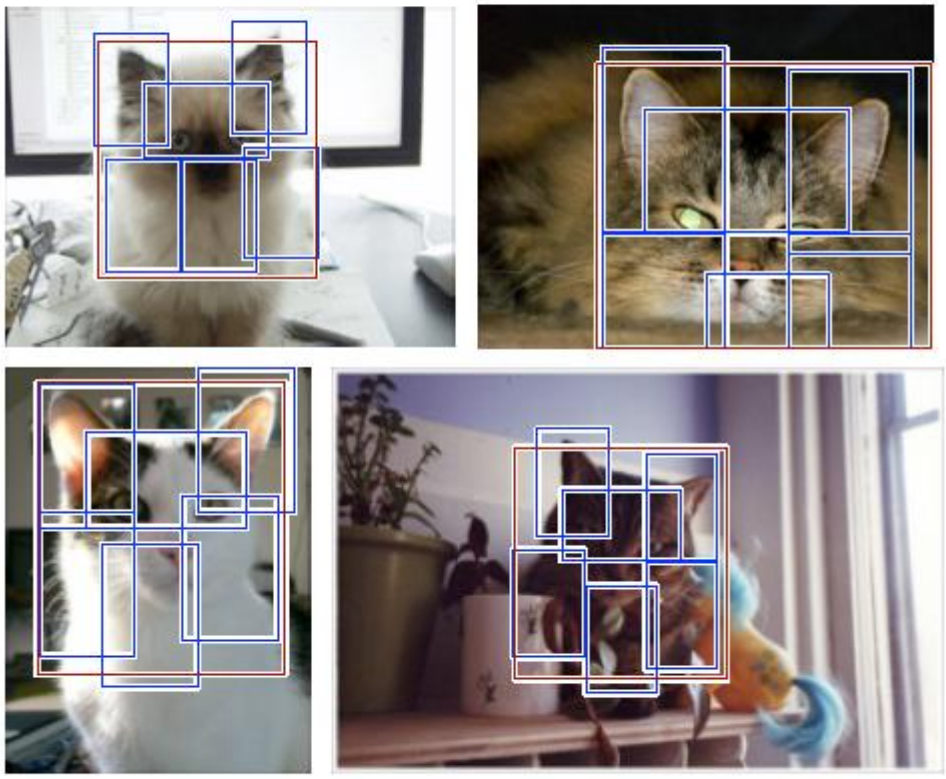


high scoring false positives

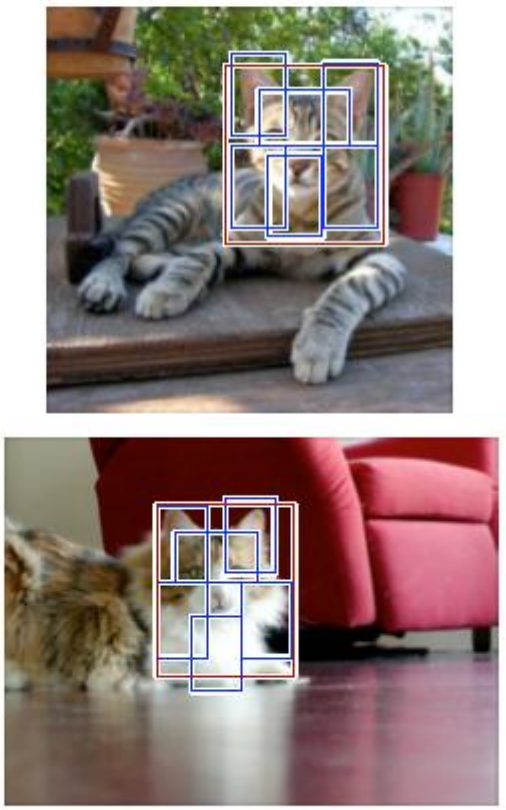


Cat detections

high scoring true positives



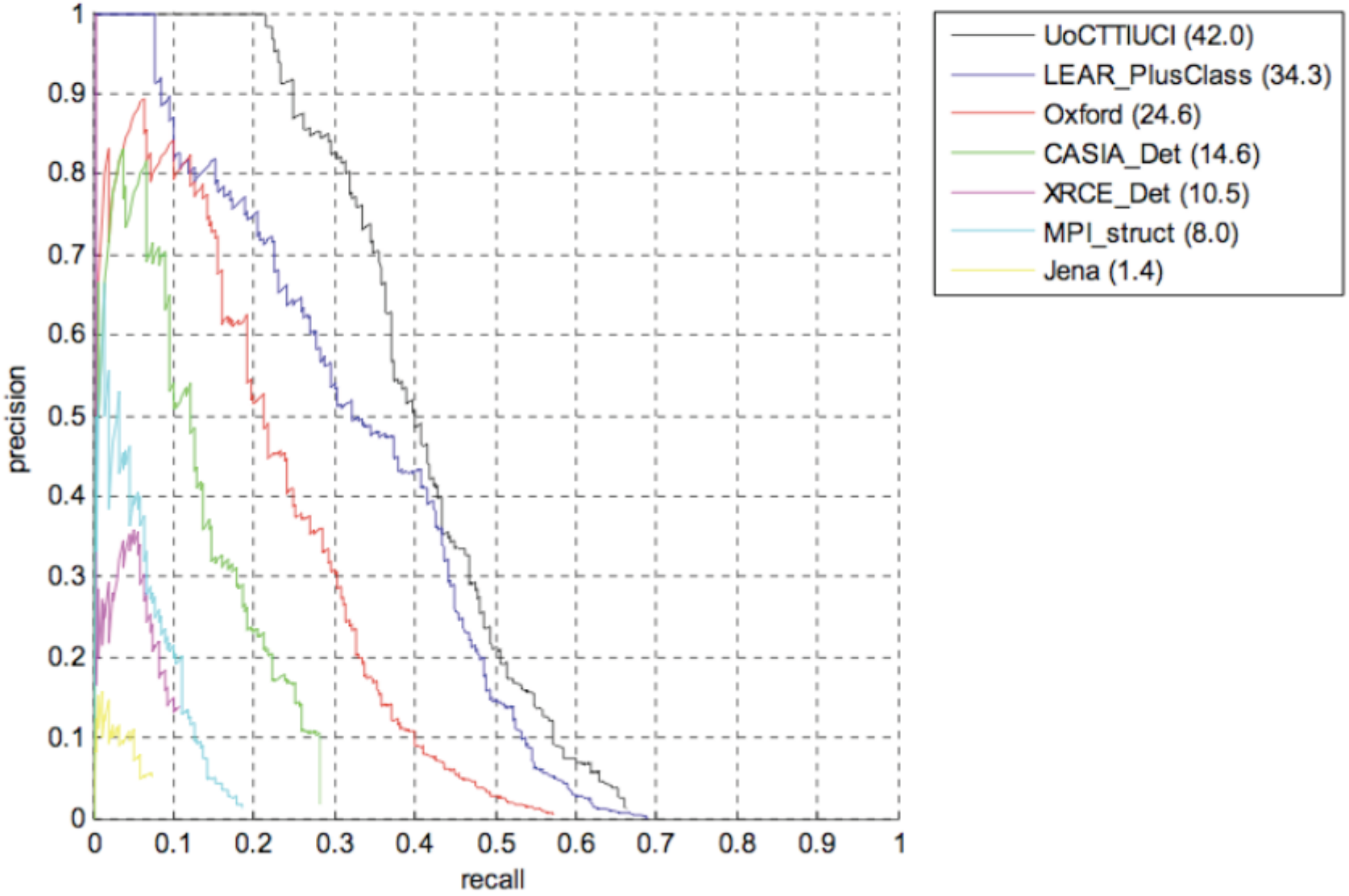
high scoring false positives
(not enough overlap)



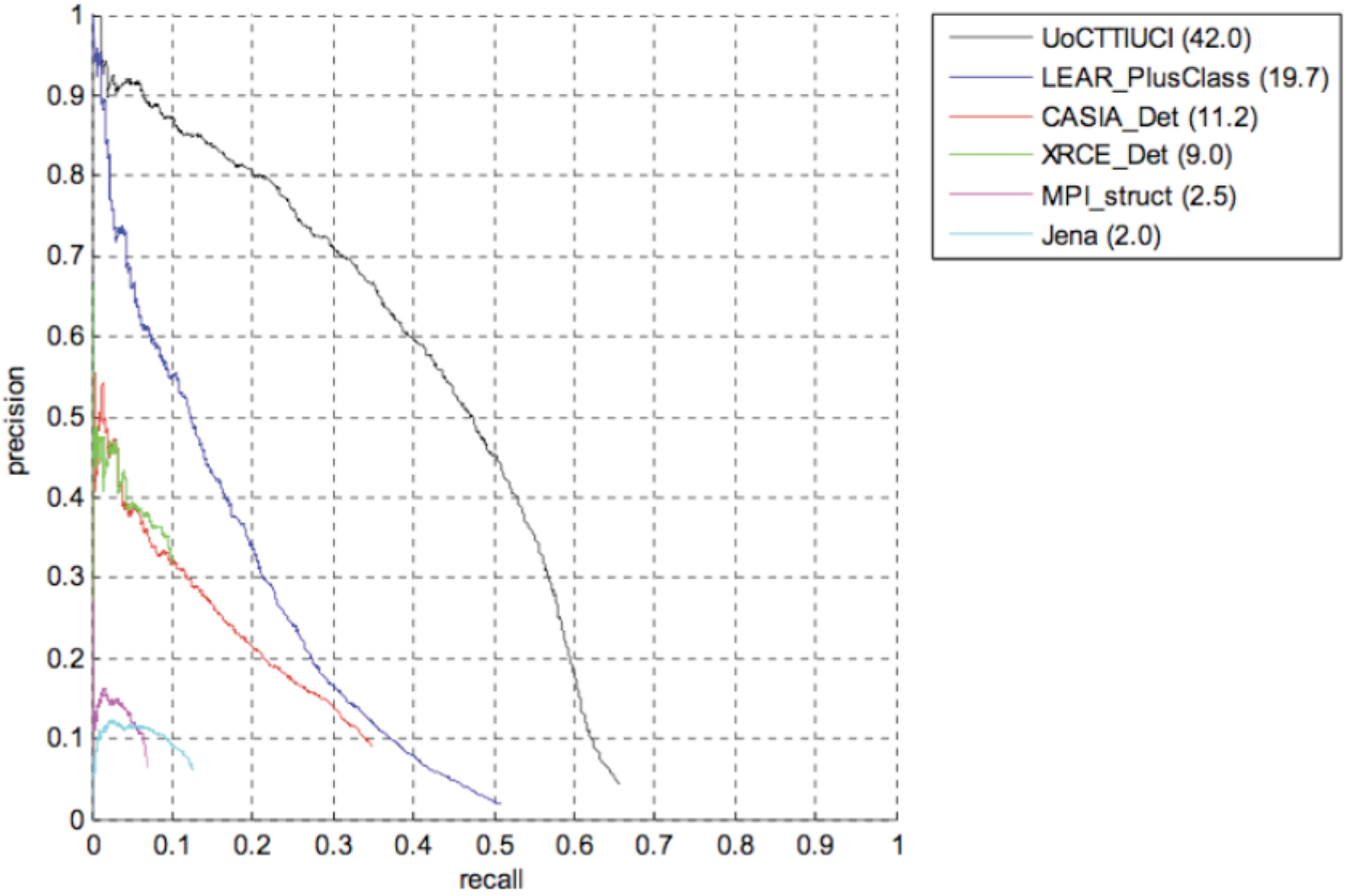
Quantitative results

- 7 systems competed in the 2008 challenge
- Out of 20 classes we got:
  - First place in 7 classes
  - Second place in 8 classes
- Some statistics:
  - It takes ~2 seconds to evaluate a model in one image
  - It takes ~4 hours to train a model
  - MUCH faster than most systems.

Precision/Recall results on Bicycles 2008

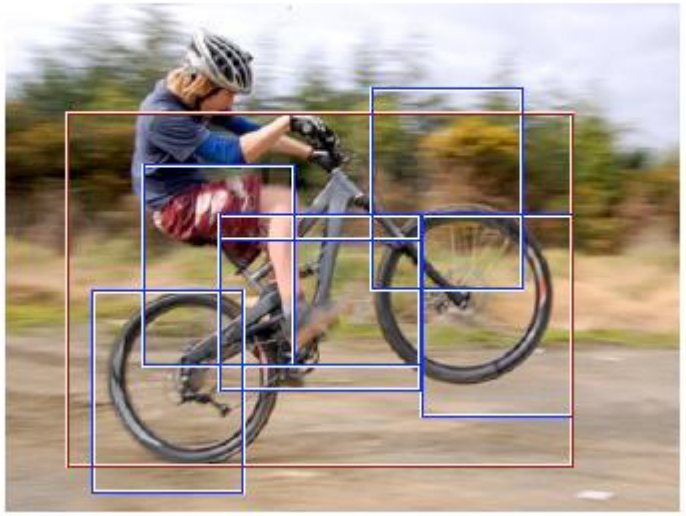


Precision/Recall results on Person 2008



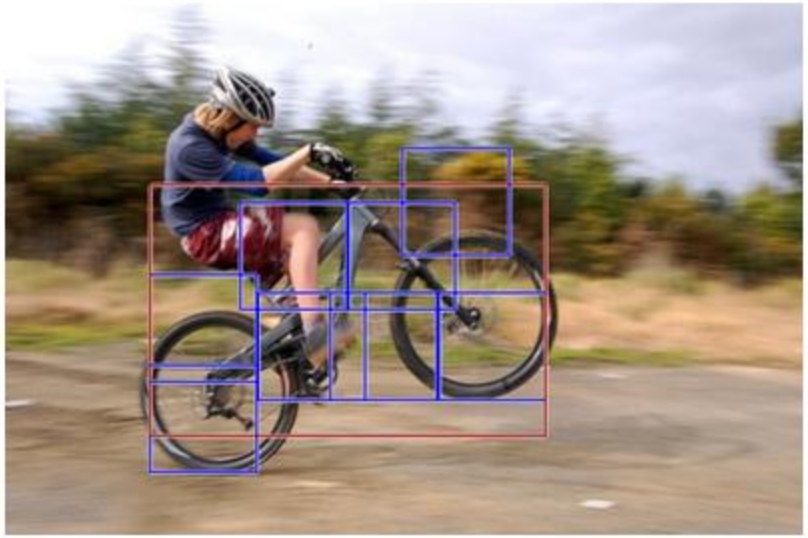
Summary

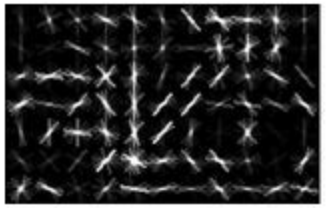
- Deformable models for object detection
  - Fast matching algorithms
  - Learning from weakly-labeled data
  - Leads to state-of-the-art results in PASCAL challenge
- Future work:
  - Hierarchical models
  - Visual grammars
  - AO* search (coarse-to-fine)

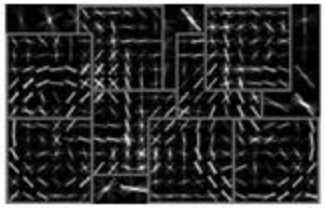


Discriminatively Trained Deformable Part Models

Version 4. Updated on April 21, 2010.









er the past few years we have developed a complete learning-based system for detecting and localizing objects in images. Our system represents objects using tures of deformable part models. These models are trained using a discriminative method that only requires bounding boxes for the objects in an image. The proach leads to efficient object detectors that achieve state of the art results on the PASCAL and INRIA person datasets.

a high level our system can be characterized by the combination of
Strong low-level features based on histograms of oriented gradients (HOG).
Efficient matching algorithms for deformable part-based models (pictorial structures).
Discriminative learning with latent variables (latent SVM).

SCAL VOC "Lifetime Achievement" Prize

The end !